

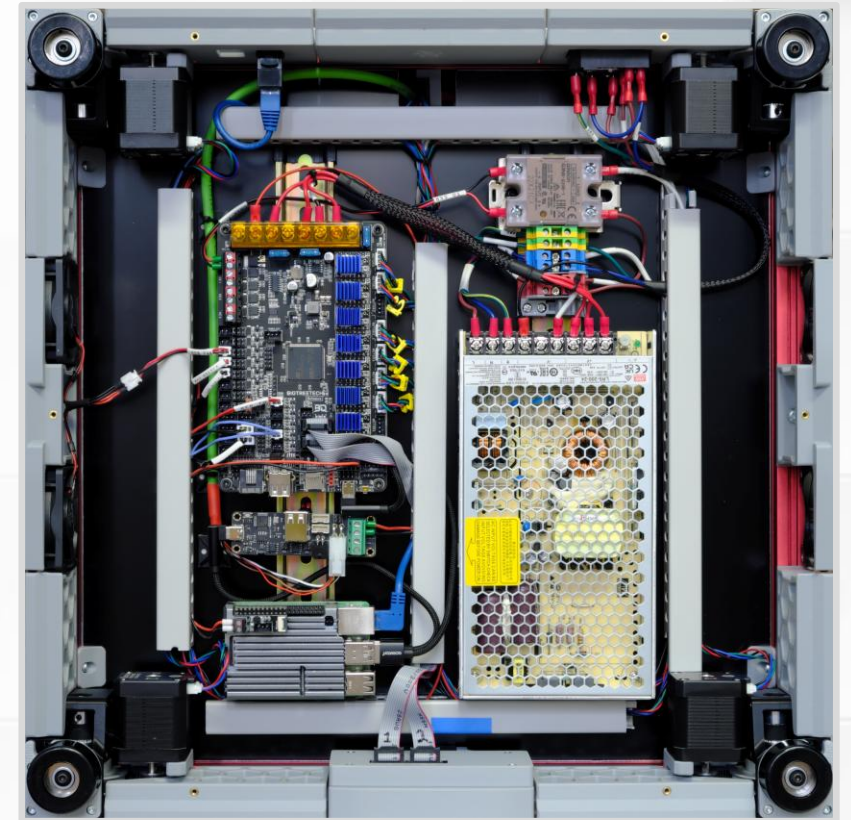
An Introduction to 3D Printing

Part 1: FFF Printers

*Process, Limitations, Dimensional Accuracy, and
Advantages*

About Me

- Hobby woodworking.
- Built my own printer to learn 3D printing.
- Voron 2.4 with over 6k hours of print time.



([Voron 2.4 GitHub](#))

Websites

- 3D designs.
printables.com/@FlyingGyroscope
- Get this guide.
github.com/FlyingGyroscope



Goals & Audience

- Training guide for students.
- Learn *strengths* and *weaknesses* of 3D printing.
- Make great prints.



Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

Thermal Expansion

3D Printing is a Process

1) Start with a 3D design.

- Check the license for restrictions when you download.
- *STL*. Approximates shapes with triangles.
- *STEP*. Real geometry. Great for 3D modeling.
- *3MF*. Compressed container, like a zip file.
Multiple objects, slicer settings, multi-color settings, ...

2) Slice and process the design with software.

- Printers do not accept 3D models!
- “Slice” objects into vertically-stacked layers.
- Create G-code file. List of toolpaths and commands that encode:
Coordinates, speeds, temperatures, ...

3D Printing is a Process

3) Print.

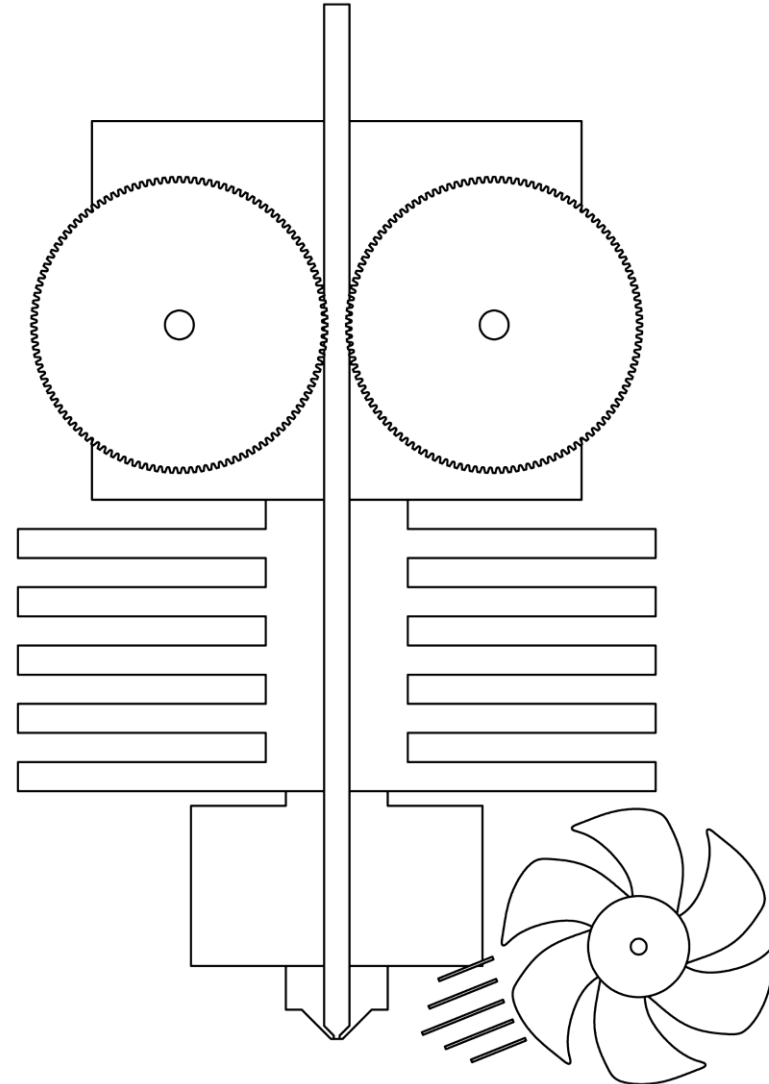
- Do the pieces fit?
- Does it work?
- Do you need to change the design? Print settings?

4) Post-process.

- Remove supports.
- Assemble.
- Glue.
- Sand or file.
- Paint.

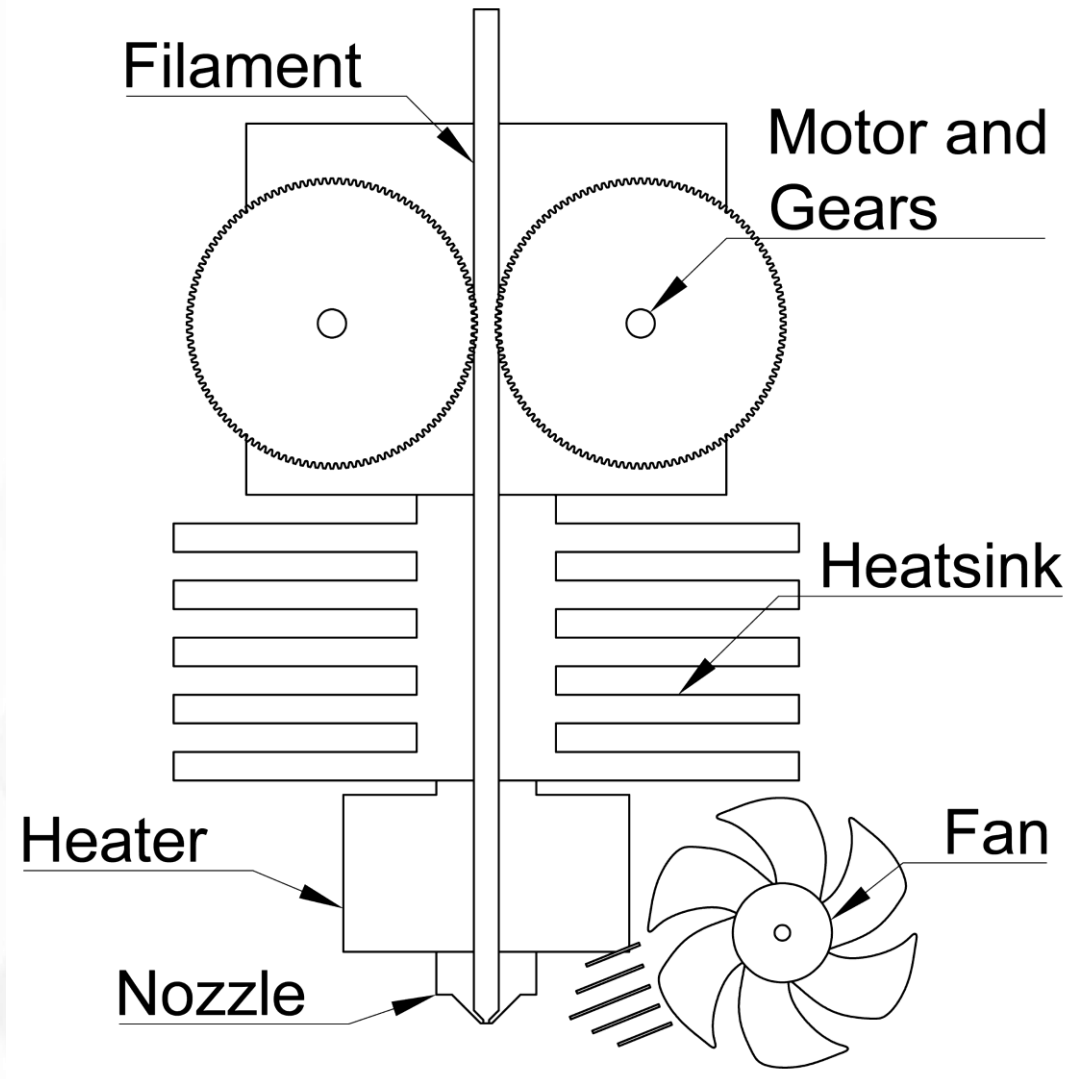
Pushing Plastic

- What kind of 3D printing?
- Filament and extrusion-based additive manufacturing.
- **FFF. Fused filament fabrication.**
- FDM™. Fused deposition modeling.
 - Essentially the same but owned by Stratasys.
- **Toolhead.** Moves around and deposits plastic.



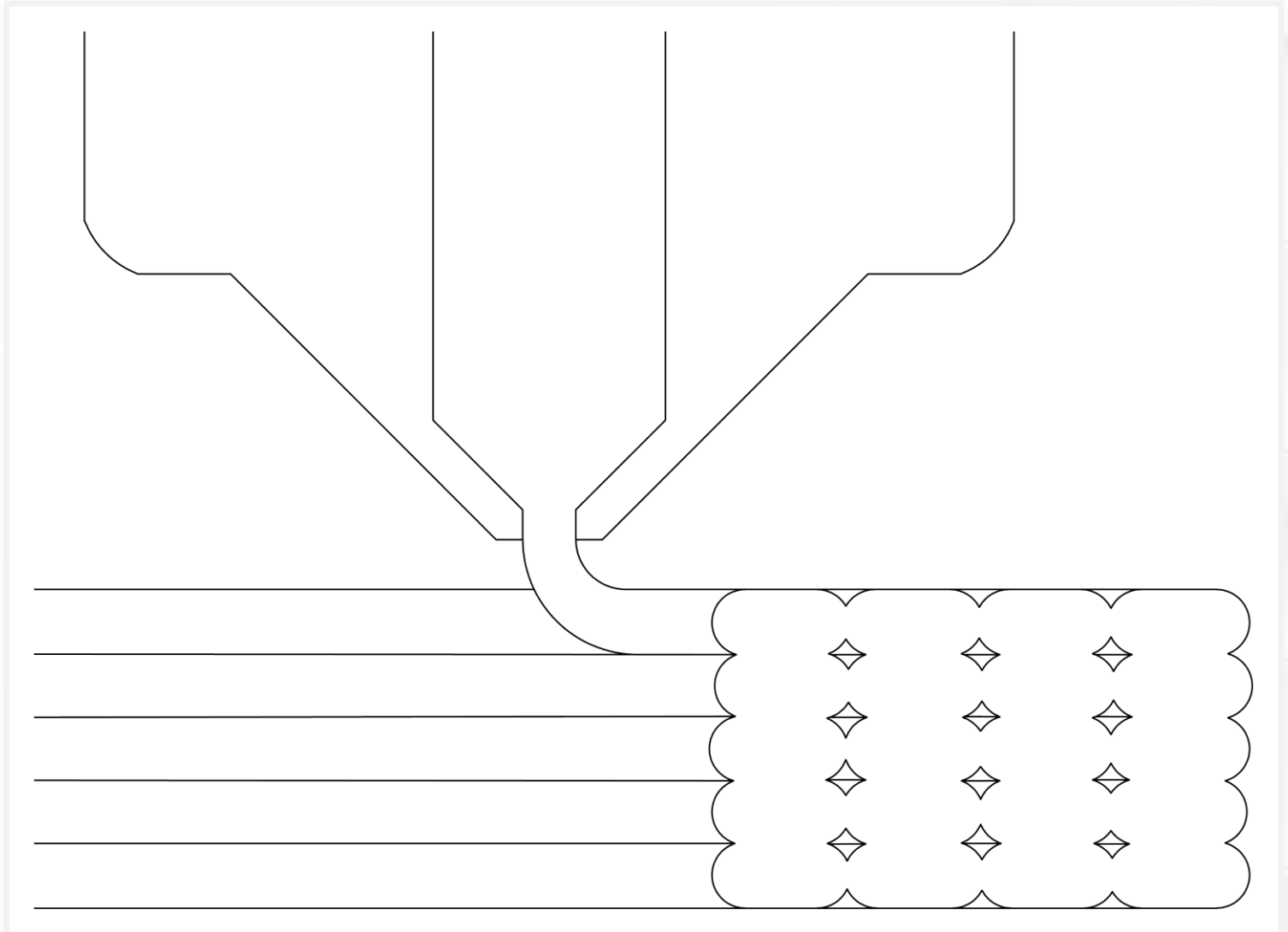
Pushing Plastic

- **Filament.** Long, continuous plastic rod.
- **Motor and gears** grab and push filament.
- **Heatsink** keeps heat in the hotend (heater + nozzle).
- **Heater** softens filament at the glass transition temperature.
- **Nozzle** opening shapes plastic.
- **Fan** cools hot plastic.



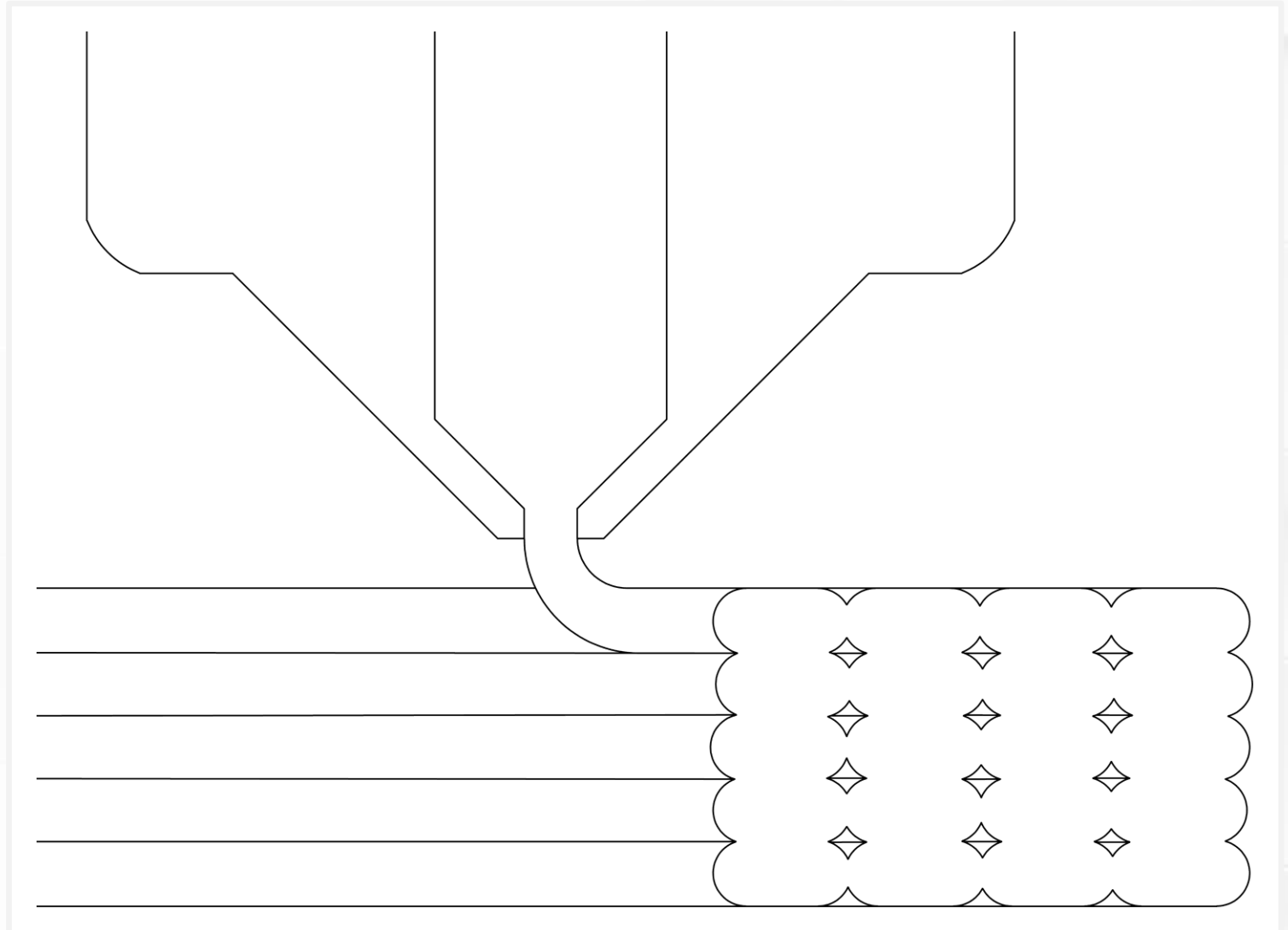
Extrusions

- Toolhead deposits lines called **extrusions**.
- Best to squish extrusions into place.
- Lock in place as they cool.



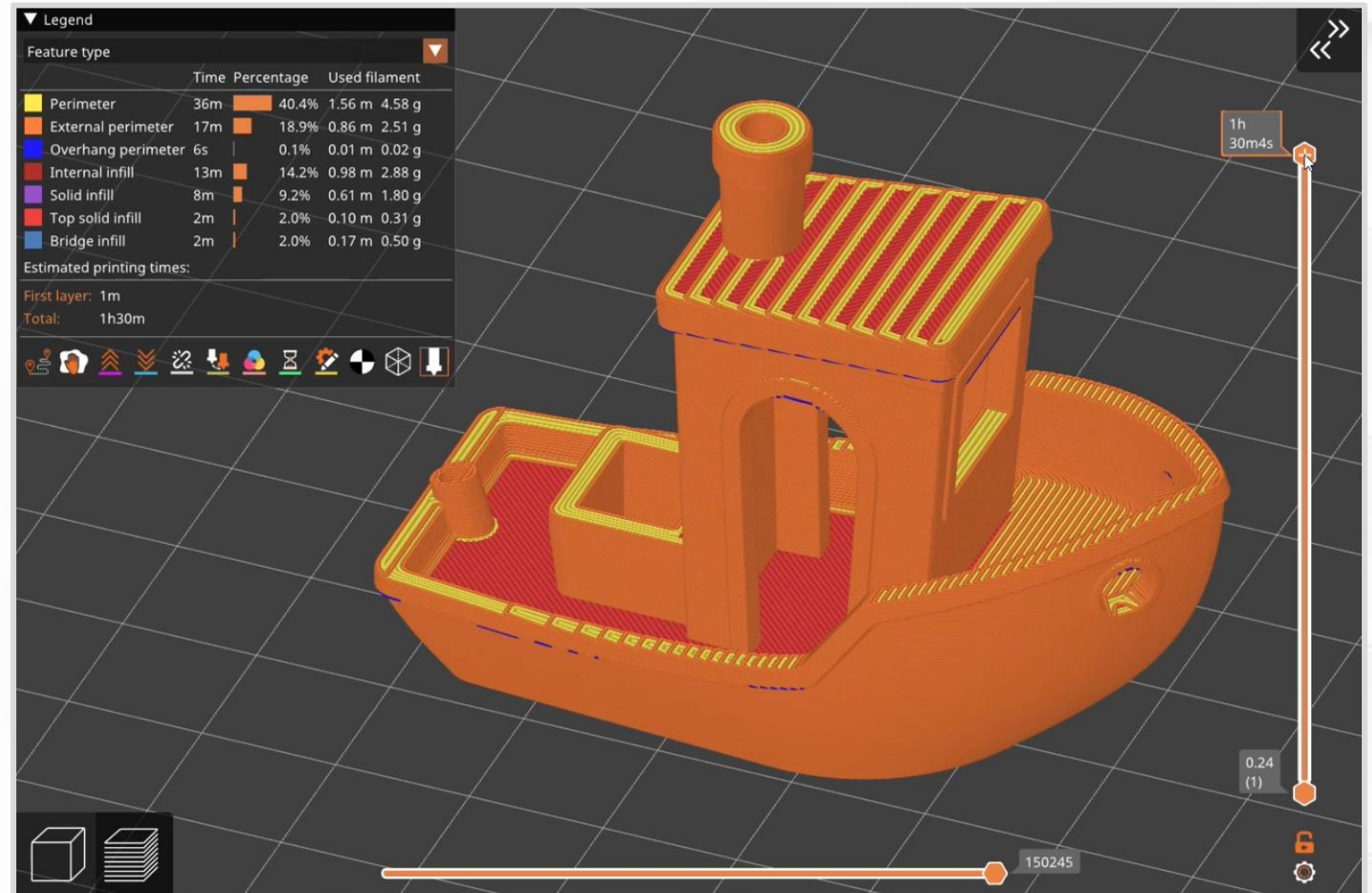
Stacking Layers

- Printers simulate 3D shapes by stacking 2D layers.
- You could average the two and call it 2.5D printing.



Slicer Layers

Let's see how PrusaSlicer handles layers.



(Animation,
FlyingGyroscope GitHub)

Slicer Layers

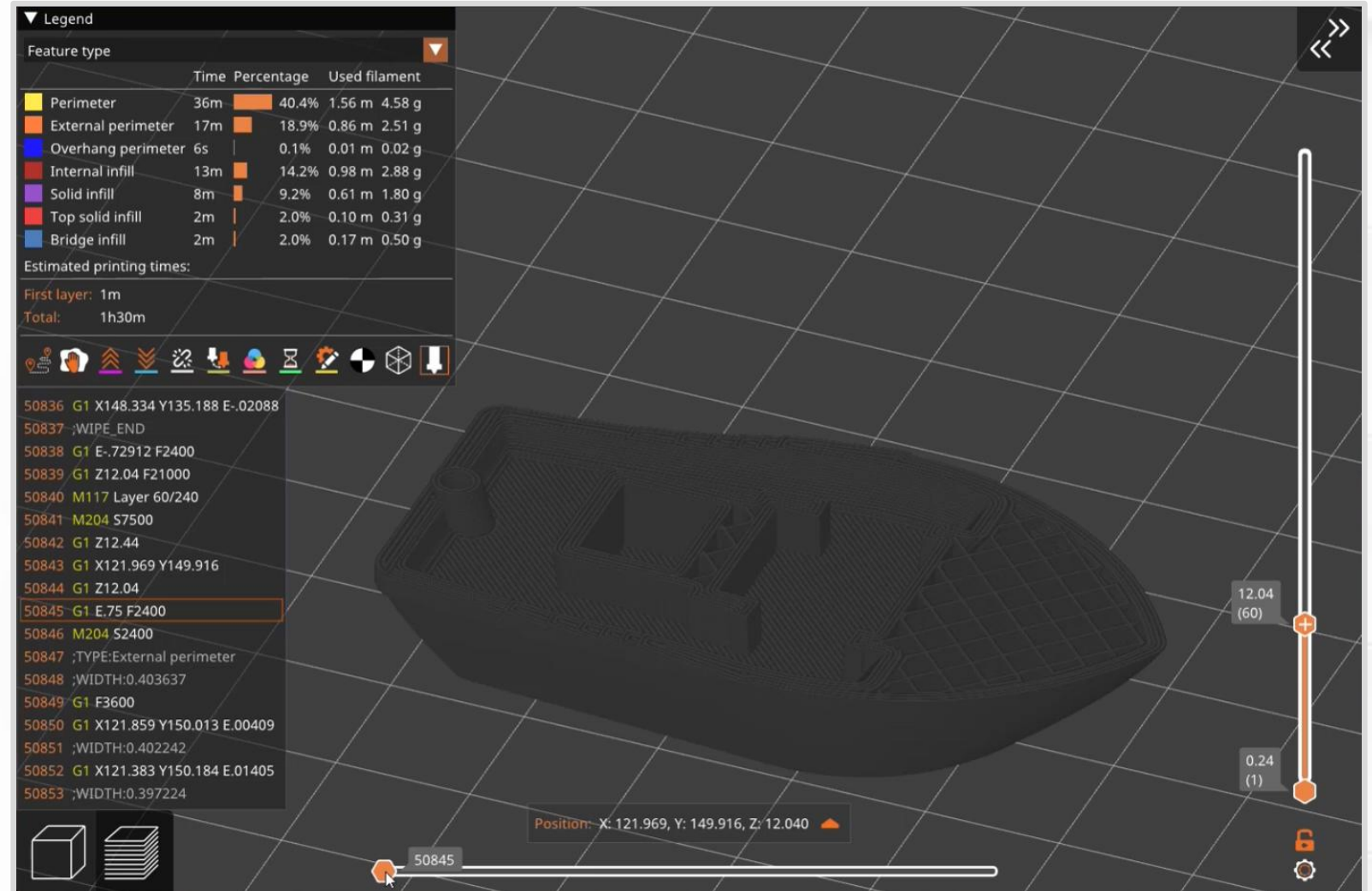
- Sort extrusions by shape and geometry.
- Color-coded.
- Adjust settings for each category. (Mostly printing speed.)

▼ Legend					
Feature type ▼					
		Time	Percentage	Used filament	
	Perimeter	36m		40.4%	1.56 m 4.58 g
	External perimeter	17m		18.9%	0.86 m 2.51 g
	Overhang perimeter	6s		0.1%	0.01 m 0.02 g
	Internal infill	13m		14.2%	0.98 m 2.88 g
	Solid infill	8m		9.2%	0.61 m 1.80 g
	Top solid infill	2m		2.0%	0.10 m 0.31 g
	Bridge infill	2m		2.0%	0.17 m 0.50 g
Estimated printing times:					
First layer: 1m					
Total: 1h30m					

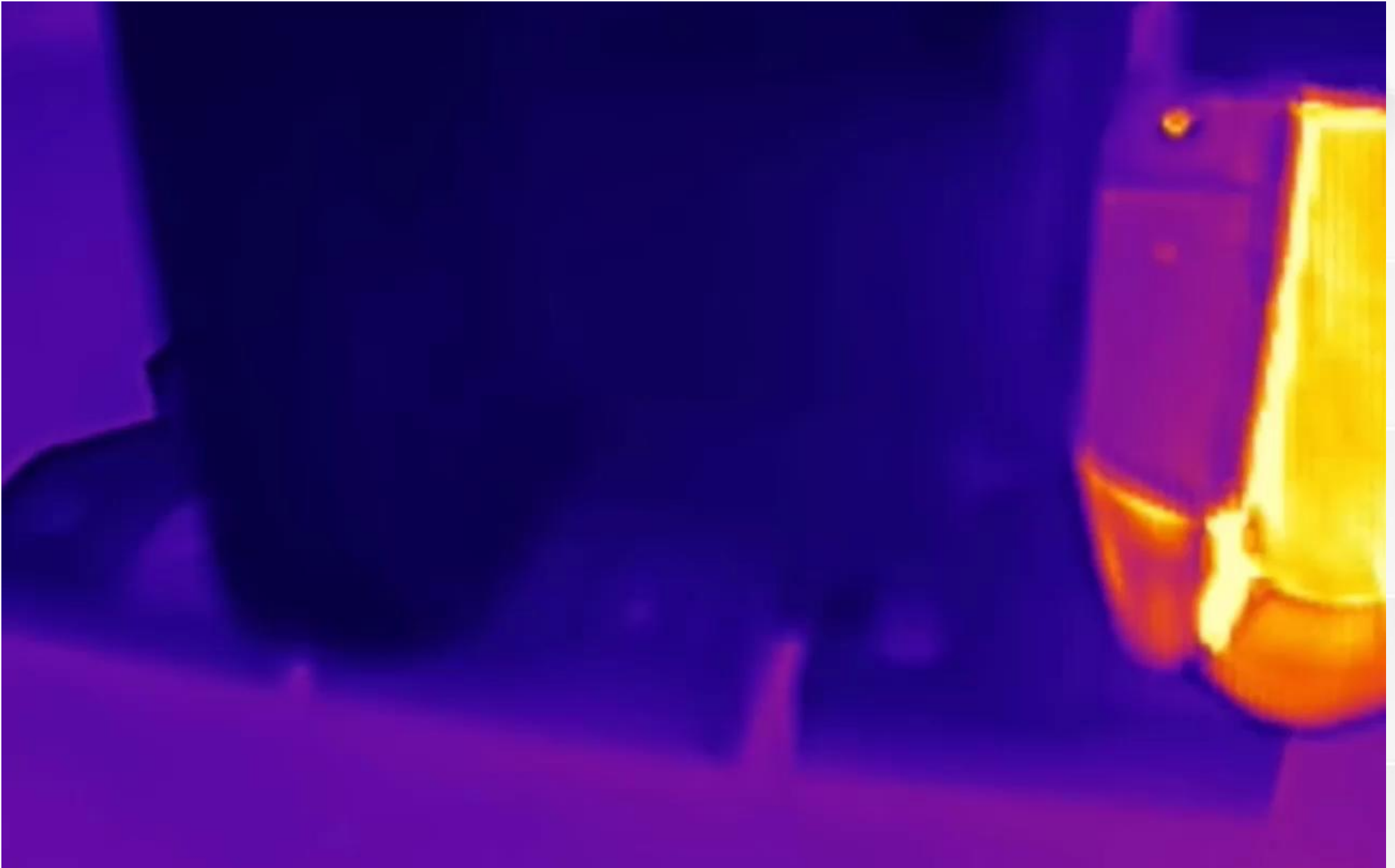
Slicer Layers

- Single layer preview.
- **G-code** shows what the printer is actually doing (lower left).
- *G1* = Move. Include X, Y, or Z coordinates.
- *G1 X Y E* = Move and extrude.

(Animation,
FlyingGyroscope GitHub)



(Animation,
Thermal camera
video by
NeedItMakelt)



Coordinating Settings

- Making consistent extrusions requires tuned settings and special techniques.
- **Hotend temperature.** Determines how easily filament flows.
 - Changes viscosity and affects all flow-related settings.
- **Part cooling fan.** Dynamically adjust cooling to hold extrusions in place.
Extra cooling for
 - Quick layers, based on minimum layer time.
 - Sagging extrusions on overhangs and bridges.
- **Extrusion Multiplier (EM, EX).** Fine tune the overall flow rate.
 - Too low = under-extruding. Gaps between lines, weak parts.
 - Too high = over-extruding. Excess material builds up, rough surfaces, offset dimensions.

Coordinating Settings

- **Maximum volumetric flow.** Stay within physical limits of hotend.
 - Prevent under-extruding at high speeds.
- **Pressure/Linear advance.** Compensate for uneven flow during changes.
 - *Example:* Pressure gradually builds at the start of an extrusion, so momentarily overshoot at the beginning.
- **Retraction.** Pull back filament and the end of an extrusion.
 - Stop flow while toolhead moves to a new location.
 - Prevent leaks, oozing, stringing, blobs.
- Check out community guides for more information.
 - [Ellis' Print Tuning Guide](#).
 - [OrcaSlicer Calibration Guide](#).

Easy Filaments

- **PLA** (derived from corn starch).
 - General-purpose filament for indoor applications.
 - Excellent bridging.
 - Very stiff.
 - Creeps. Deforms permanently under load.
 - Low temperature resistance. Softens inside a car on a hot summer day.
- **PETG** (plastic water bottles).
 - Pretty good general-purpose filament.
 - Hygroscopic. Needs low-humidity storage.
 - Elastic. Excellent choice for springs.
 - Moderate temperature resistance. Can be used outdoors.

Easy Filaments

- What makes PLA and PETG easy?
 - Low warp. Low shrink.
 - High dimensional accuracy.
 - Room temperature, open-air (non-enclosed) print chamber.

Proper Protocol

Let's review some everyday guidelines.

- Secure the loose end of a spool of filament at all times.
 - Tangles ruin prints and waste time.
- Do not reach into the printer while it is running.
 - Use touchscreen or buttons to pause and resume prints.
- Watch the first layer before you walk away.
- Avoid burns.
 - Read nozzle and print bed temperatures on the display.
 - Never reach near a hot nozzle (210 °C = 410 °F).
- Avoid leaving fingerprints on print sheets.
 - Clean sheets as they get dirty.

Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

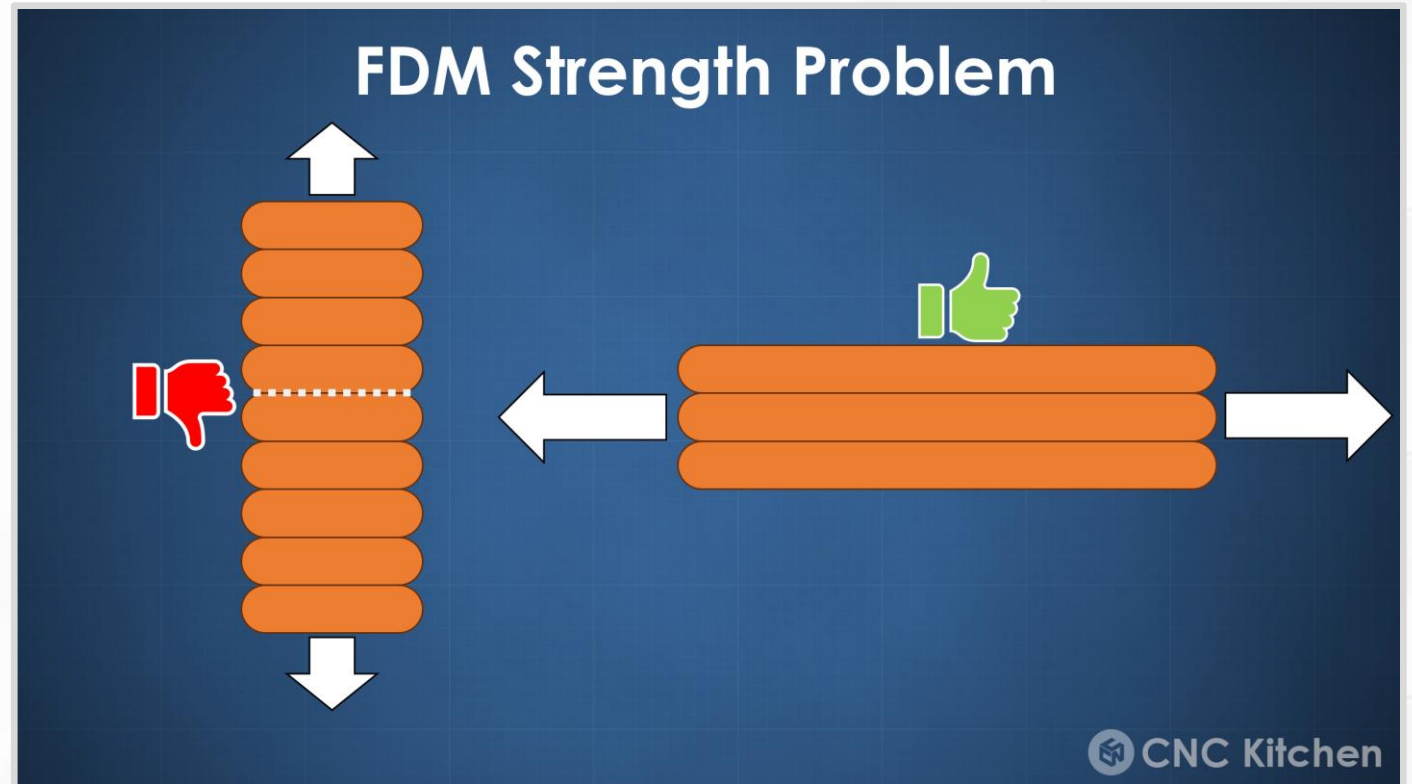
Thermal Expansion

Thermal Cycle

- Some plastics **shrink** as they cool.
 - Up to 1 or 2%.
 - Seems small, but large enough to distort dimensions.
 - Leads to cracking and delamination in extreme cases.
- Advice?
 - Tightly control ambient temperature and cooling rate.
 - Some filaments (ABS, Nylon) like it hot → Enclosed print chamber, extra heating.
 - Some filaments (PLA) like it cold → Open-air printing, extra cooling.
 - PETG is somewhere in the middle.

Anisotropic Strength

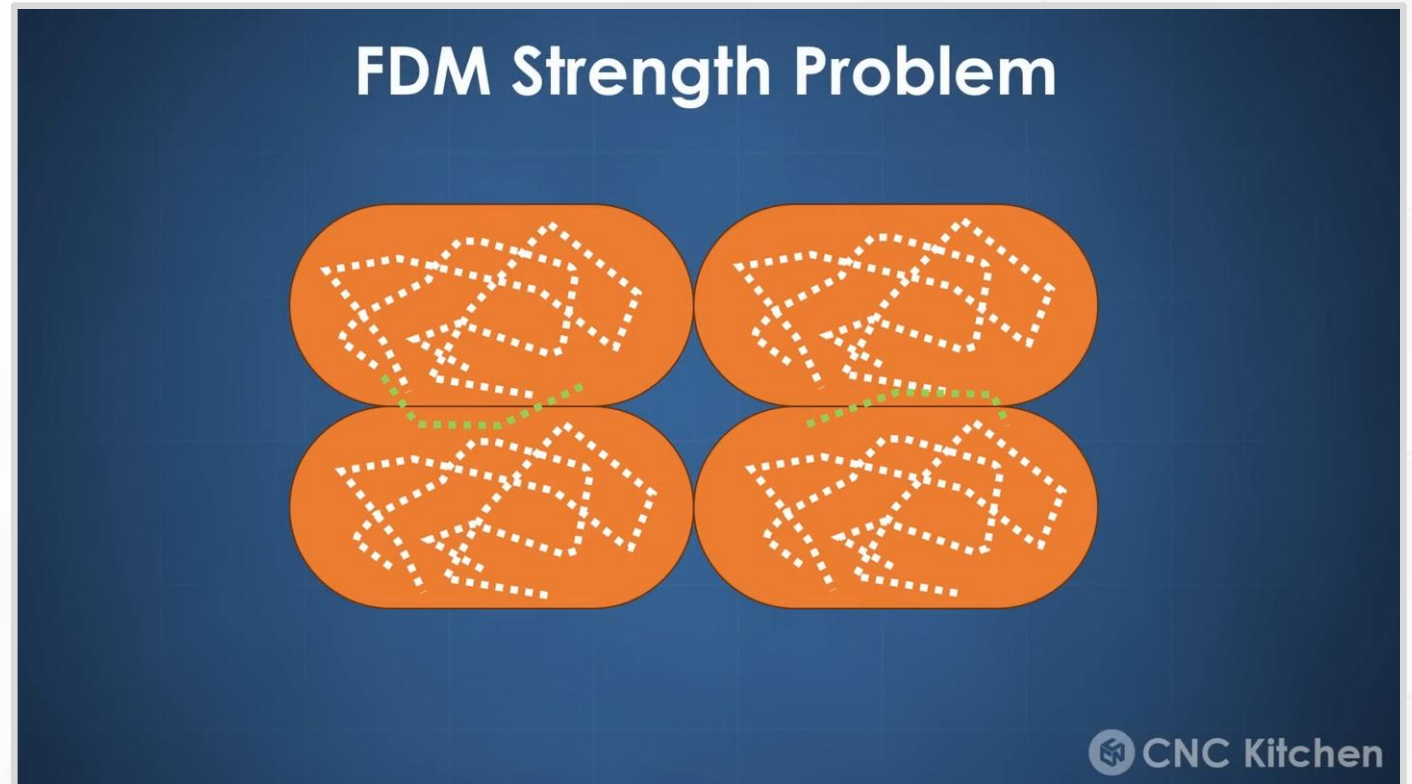
- Print orientation and layer lines affect strength.
- Layer-to-layer bonding is a weak point.
- And all of those weak points align.



([YouTube video](#) by CNC Kitchen)

Anisotropic Strength

- Within an extrusion, hot filament bonds to hot filament.
- For a new layer, hot filament partially bonds to a cold layer.
- The “wrong” print orientation can cost around 50% of part strength.



(YouTube video by CNC Kitchen)

Anisotropic Strength

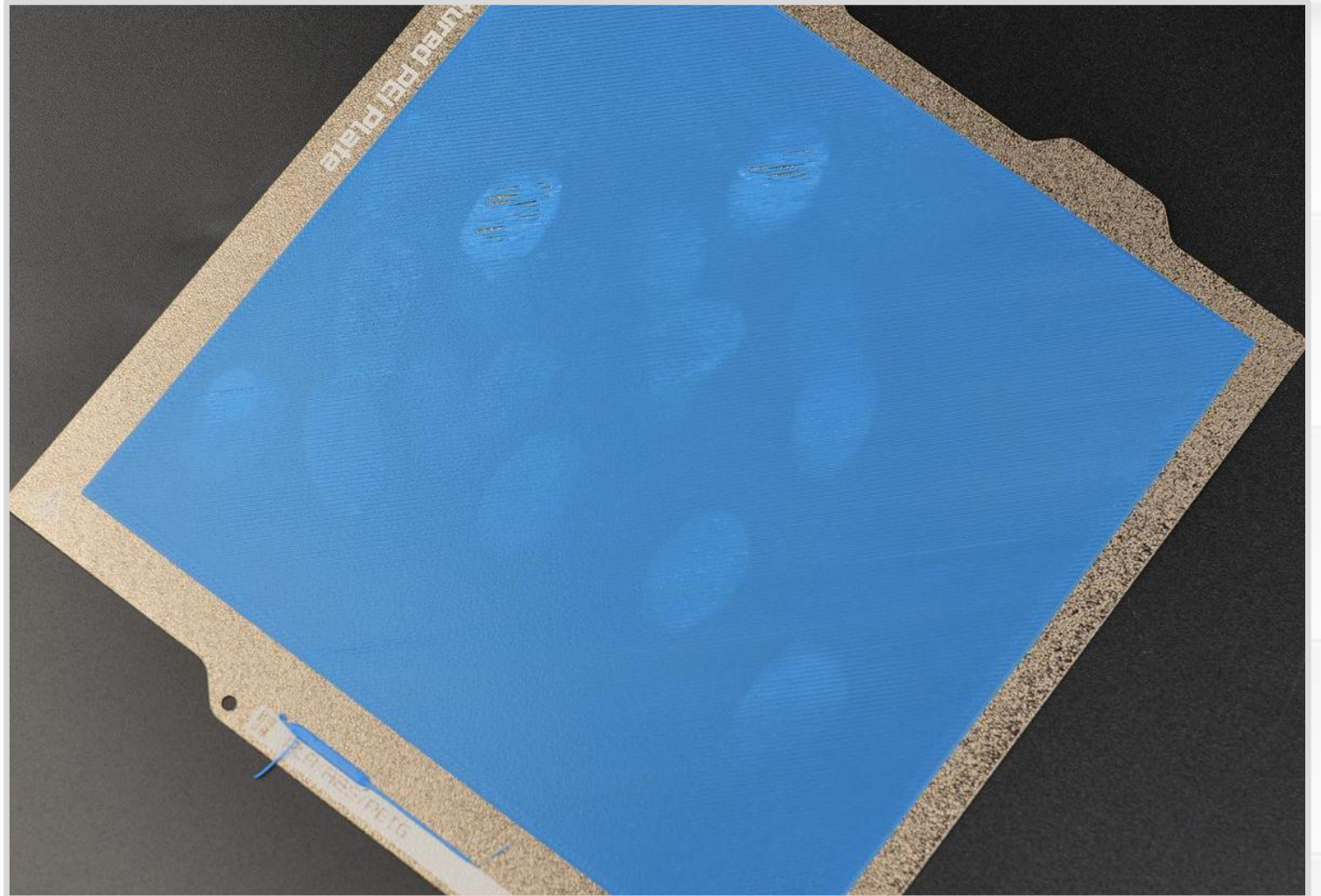
- Advice?
- Rotate prints to align layer lines and forces.
- Avoid tall and skinny shapes.
- Improve layer bonding.
 - Increase printing temps.
 - Decrease printing speed.
 - Alter design or slicer settings to create more surface area for layer-to-layer bonding.

First Layer

- What is the most important layer in a 3D print?
 - **The first layer!**
 - First layers keep everything locked in place.
- Advice?
 - Always watch the first layer.
 - Tune first layer squish (extremely sensitive).
 - Flat surface area on the bottom of the 3D object.
 - Heated print bed.
 - Perfectly clean print bed.
 - Apply adhesion promoters (glue).

First Layer Mistakes

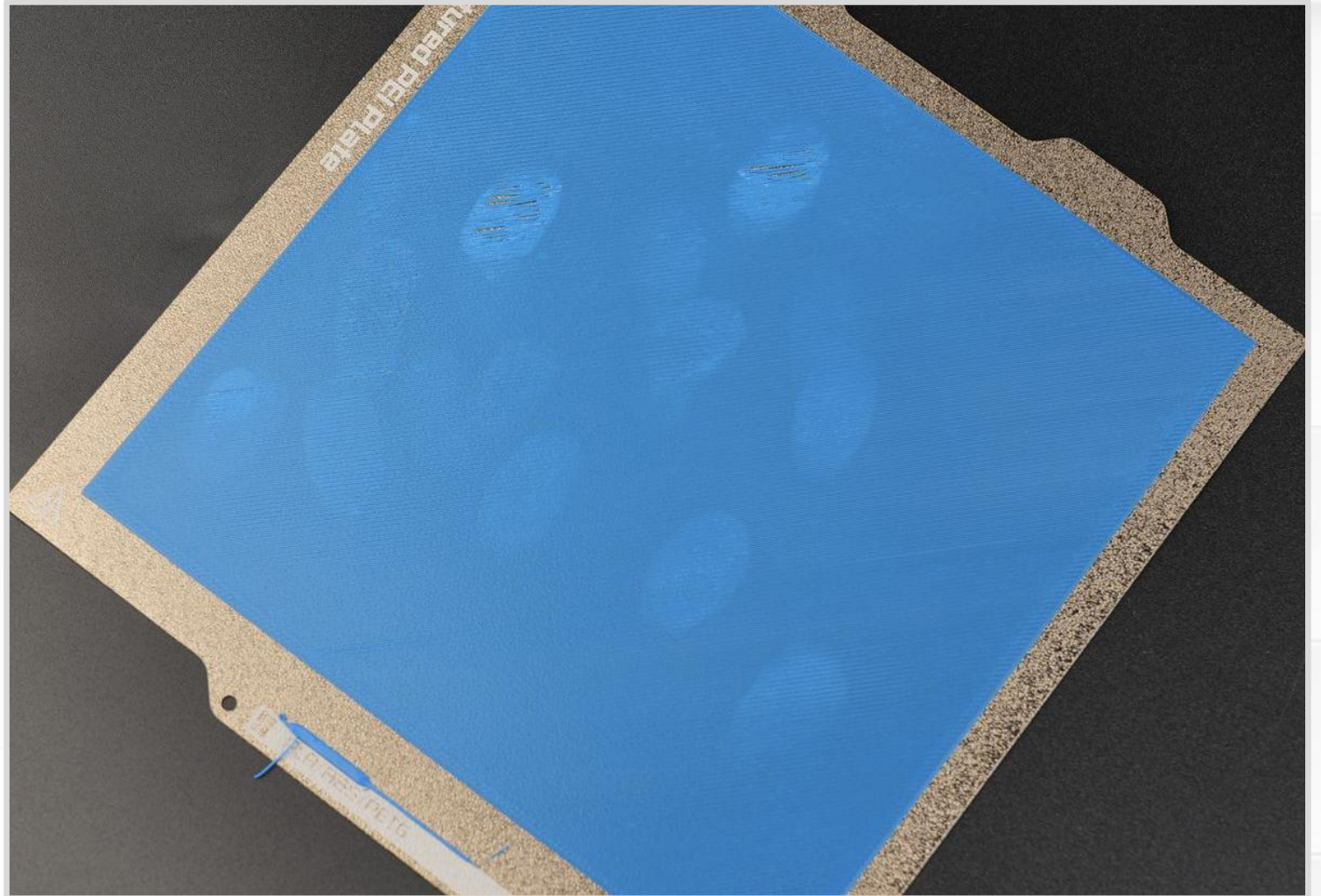
- What is wrong with this picture?
- **Fingerprints!** Plastic does not stick to oily fingerprints.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

- Advice?
- Deep cleaning: dish soap, soft brush, and water at the sink.
- Everyday use: window cleaner with ammonia.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

- What happens when a first layer does not stick?
- Print breaks free and makes “spaghetti”.
- When did things go wrong?
- Look at the partially completed, rectangular print.



(Screenshot from Everything STEM)

Stair-Stepping

- Printing with layers → Hills and vertical slopes turn into stair steps.
- Mostly cosmetic effect.
- More pronounced on shallow angles.
- Advice?
- Rotate objects to hide layer lines.
- Use smaller layer heights at the cost of increased print time.



(Picture from Hamburg University Research Project)

First Layer Warping

- Why do corners and edges peel up?
- Regions shrink at different rates → Internal stresses → Stresses concentrate in corners.
- Shrinking is a percent → Large parts have the most problems.



(Picture from [Creality Cloud Blog](#))

First Layer Warping

Consider this analogy.

- Imagine you want to remove a label or piece of tape.
- What do you do?
- Start peeling up the corner because that is the weak point.
- Same thing here.
- Physics found the weak point for you.



(Picture from [Creality Cloud Blog](#))

First Layer Warping

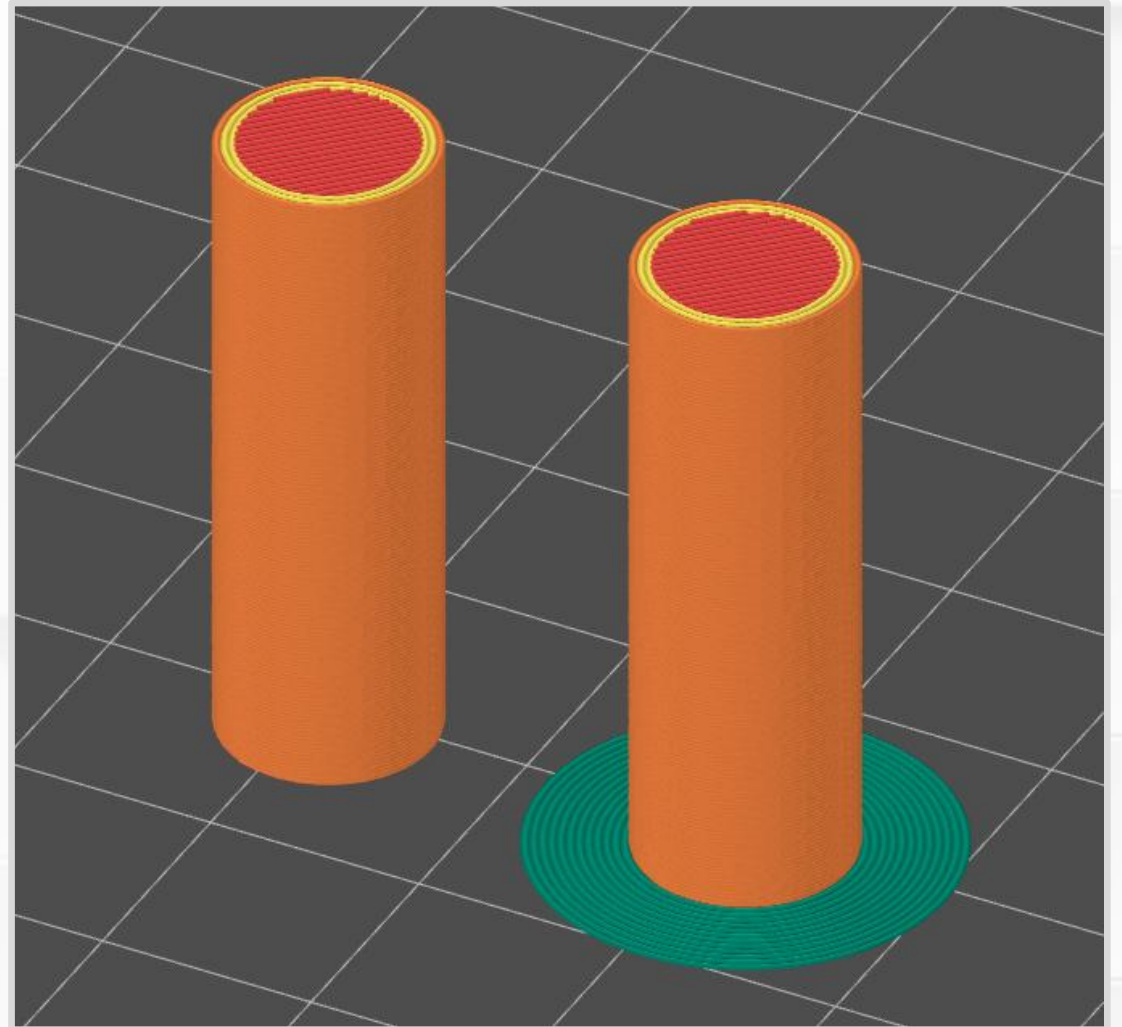
- Advice?
- Better control over temperatures.
 - Filament temperature.
 - Print chamber temperature.
 - Cooling fans.
- Evenly distributed cooling.
 - Avoid fans or drafts that cool one side.
- Round bottom corners to reduce stress concentration.



(Picture from [Crealitty Cloud Blog](#))

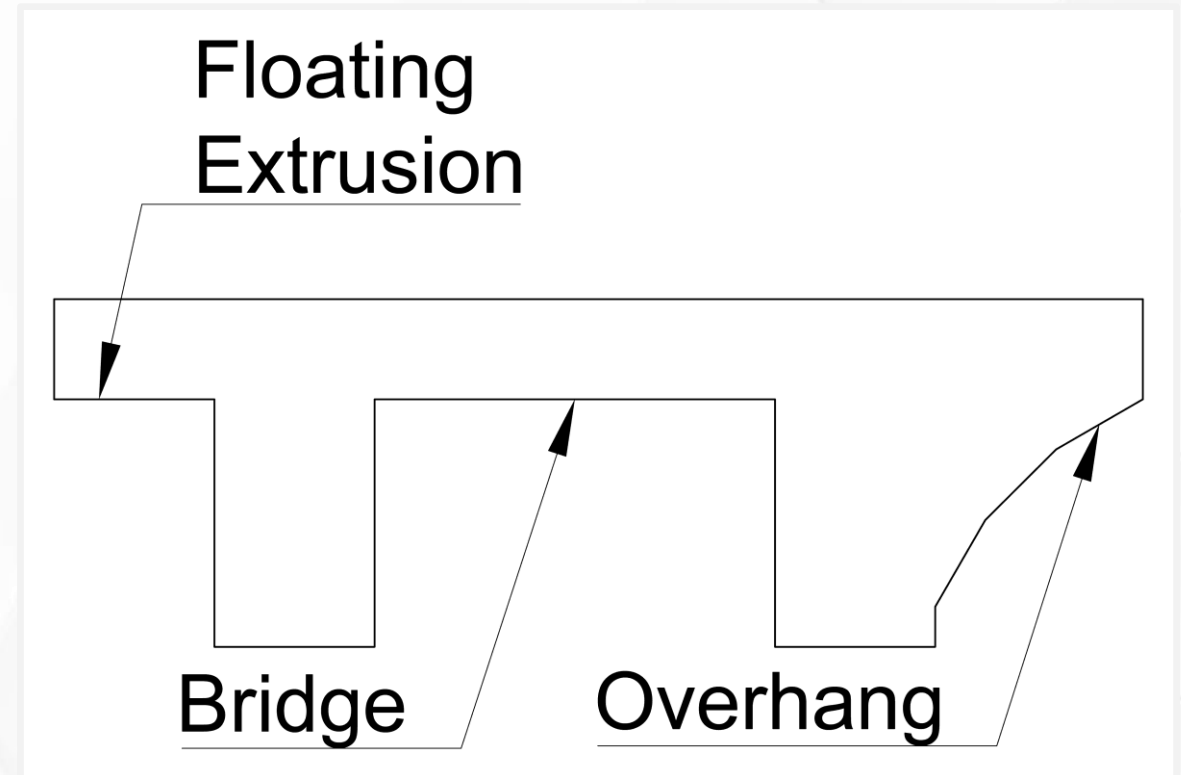
Brim

- What if you could add extra surface area to fight warping?
- Brim, like the brim on a hat.
- Concentric extrusions joined to bottom perimeter. (Green lines.)
- Added with a slicer.
- One layer tall.
- Easily peeled away after printing.



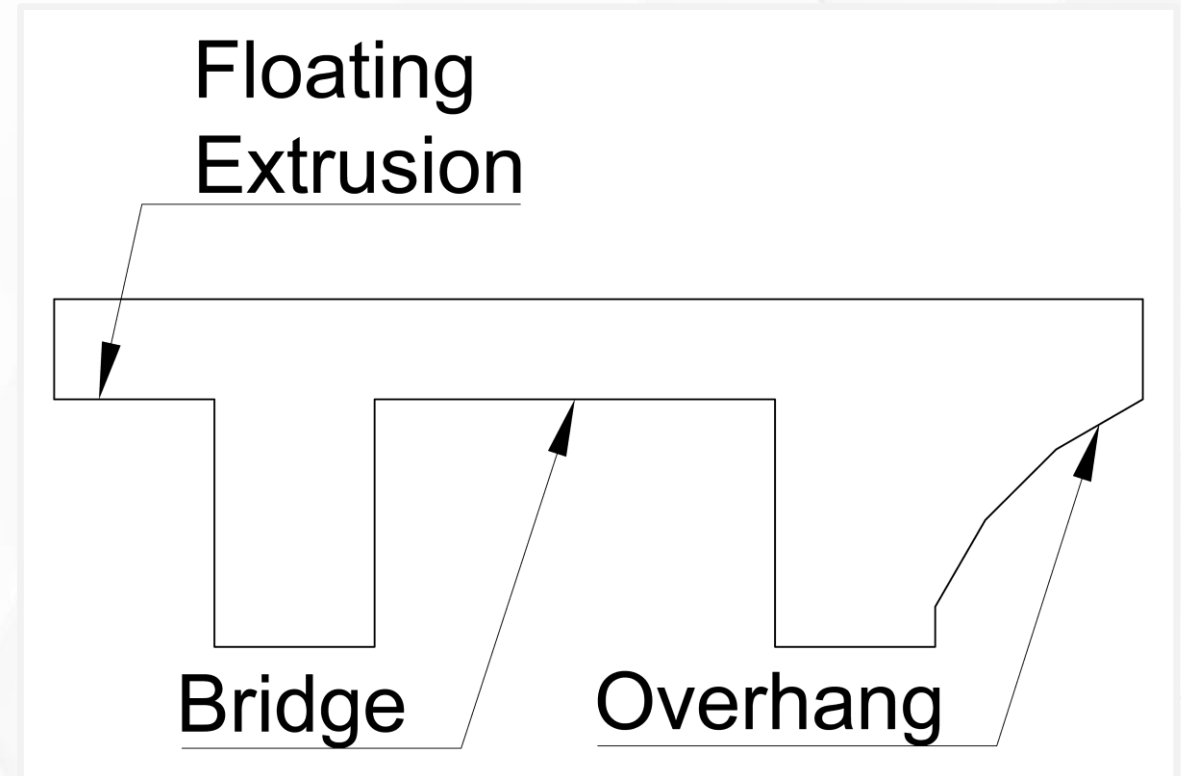
Levitating Extrusions

- *Critical thinking:* What happens when you try to levitate extrusions?
- Extrusions work best when you press them onto a solid surface.
- Predict what this print will look like.
- What parts will be the most deformed?



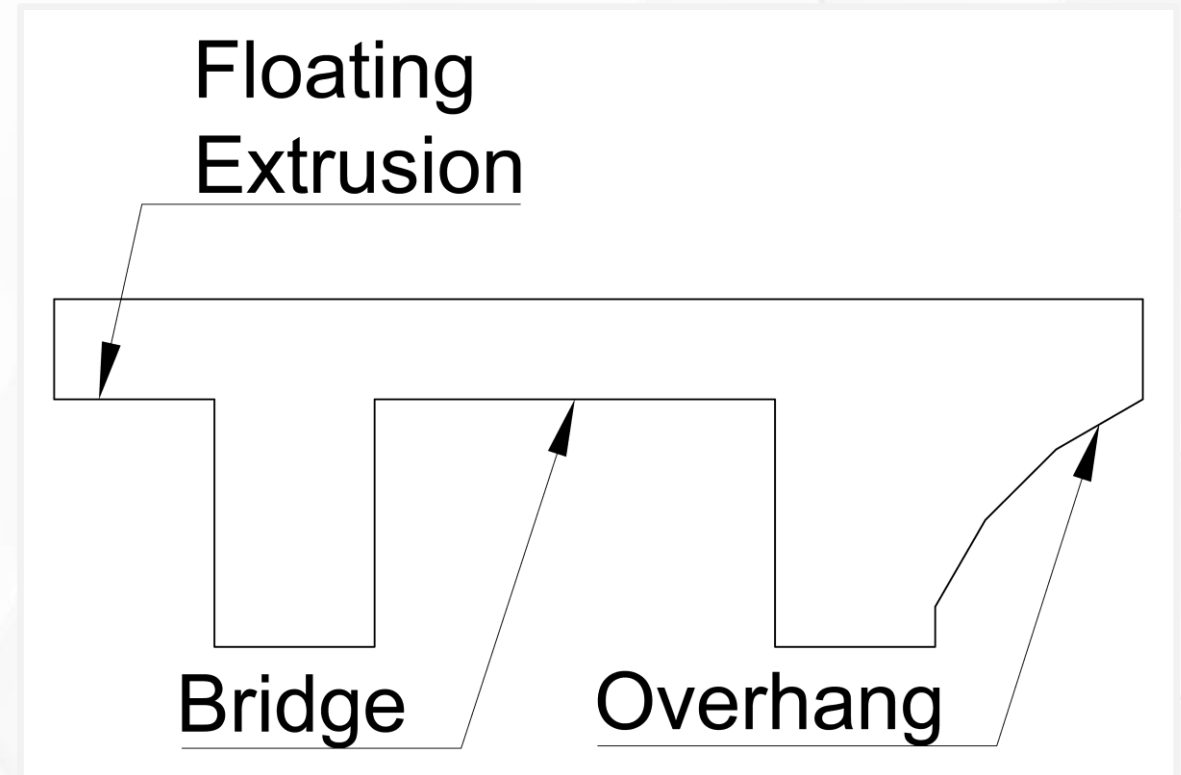
Overhang

- Extrusions that lean over an edge.
- Only the outermost extrusion is affected.
- Partially supported along one side.
- Advice?
- Keep overhangs at 45° to be safe.



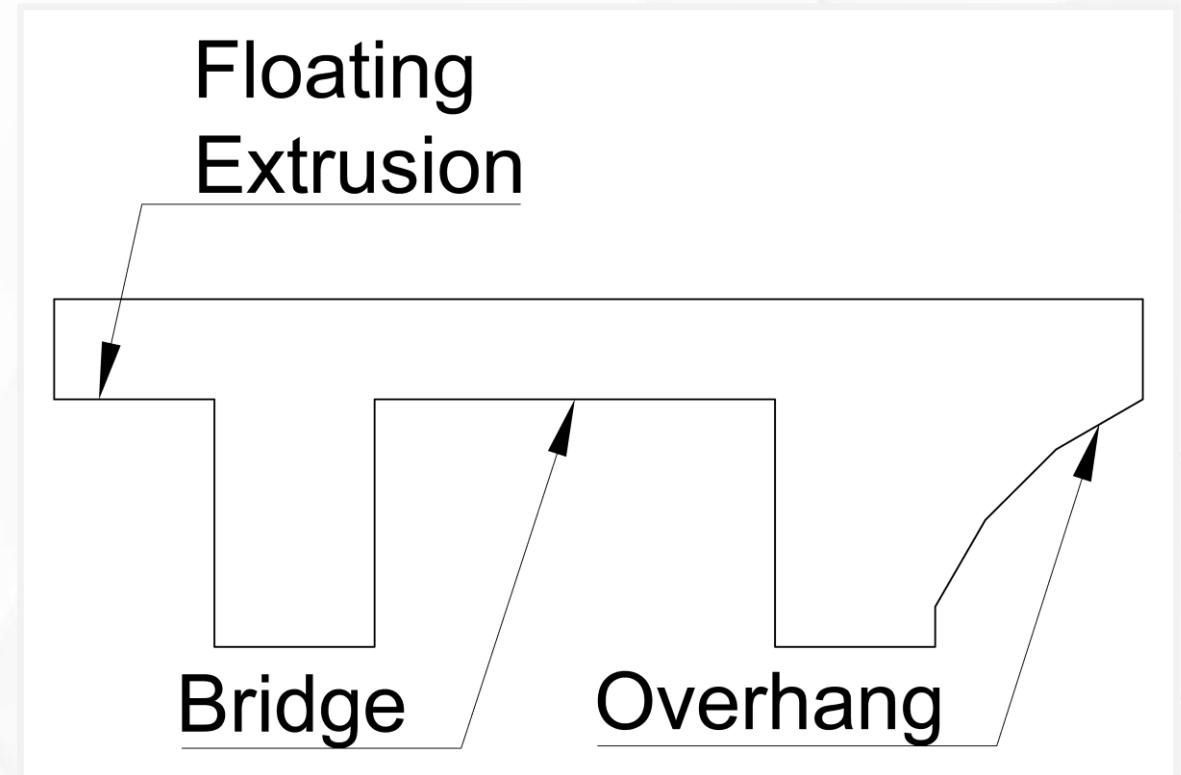
Bridge

- Both ends of an extrusion are supported and anchored.
- Middle hovers over a gap.
- First few layers sag and droop down.
- Eventually recovers to flat layers.
- Advice?
- Keep bridges as short as possible.



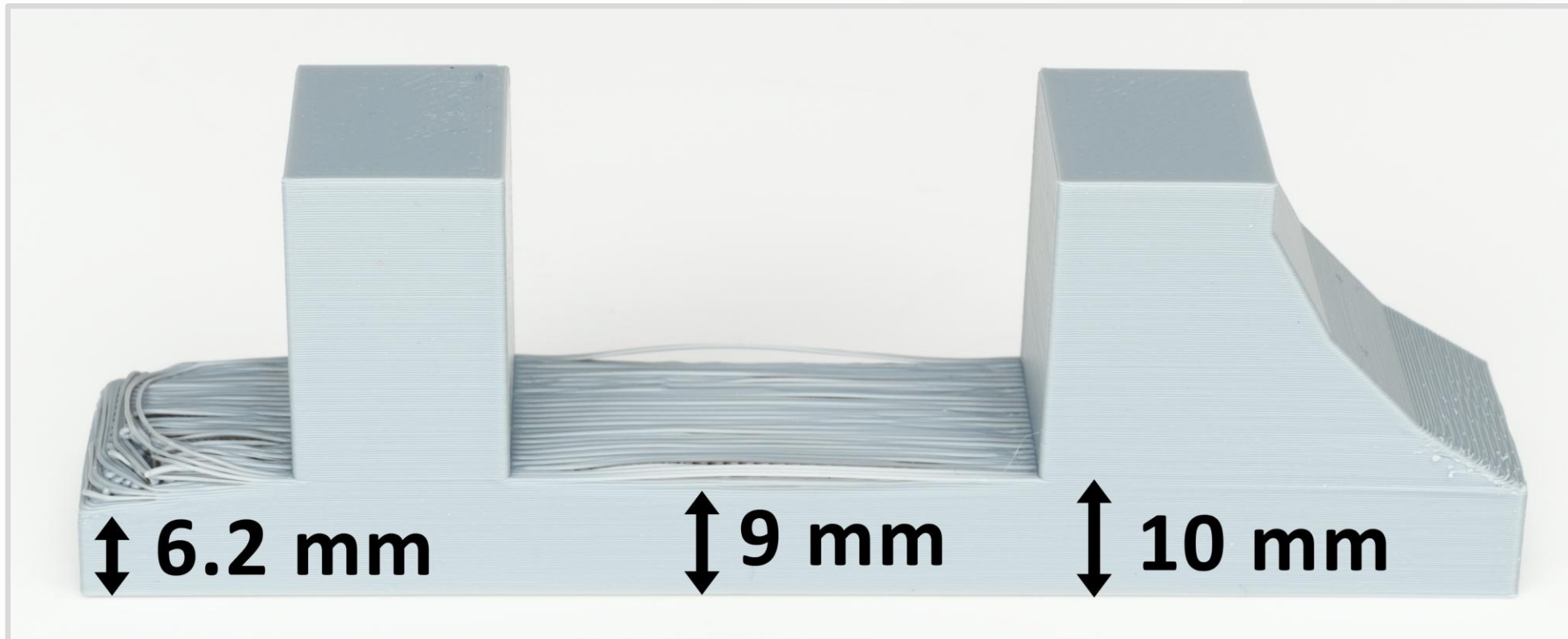
Floating Extrusion

- Extrusion that hangs in the air.
- At least one end is not anchored to a solid surface.
- Advice?
- **Avoid. Very unreliable.**



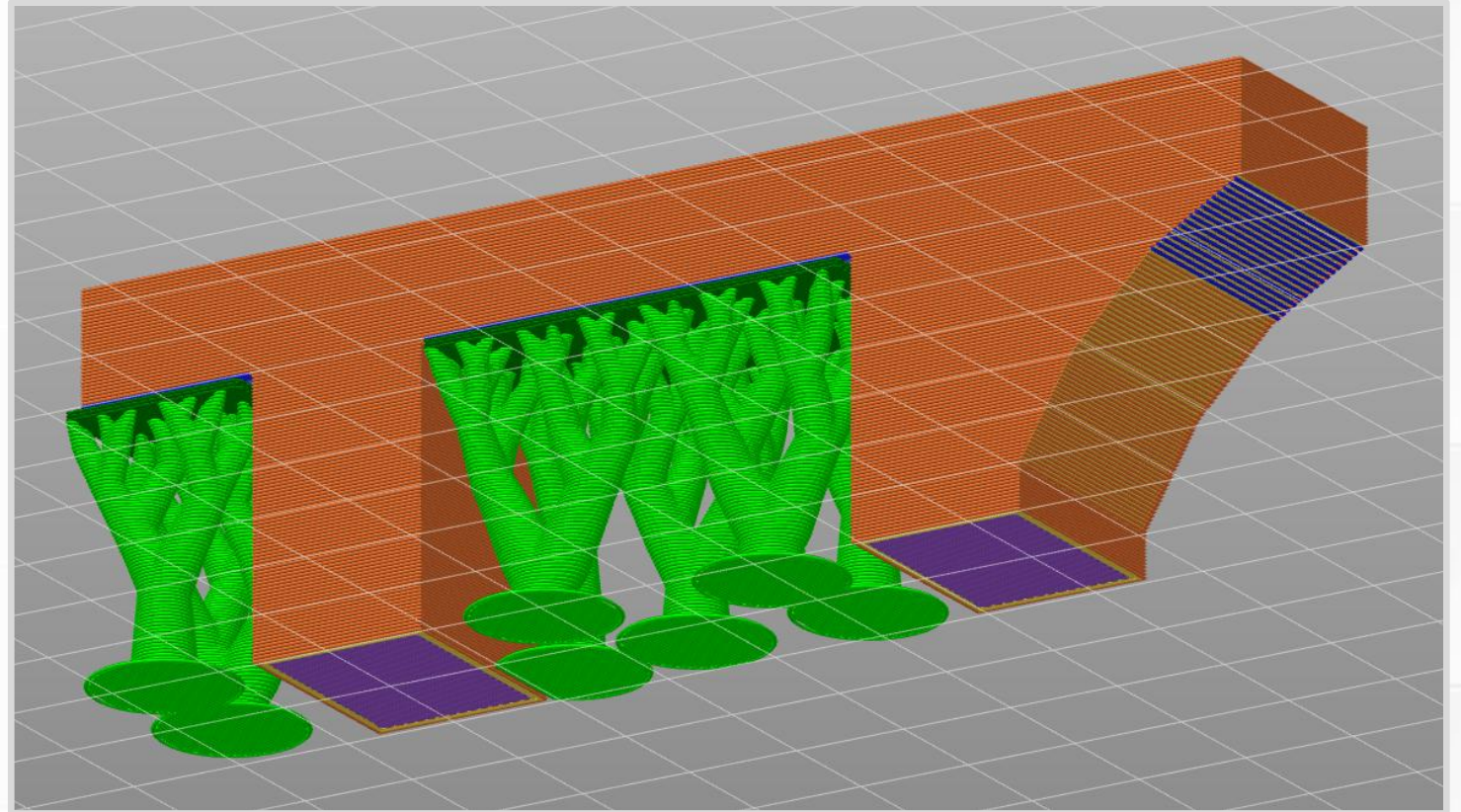
Stress Test Results

- All overhang angles look good.
- Long bridge recovers to flat layers in 1 mm (**5 layers**).
- Short floating extrusions recover in 3.8 mm (**19 layers**).



Supports

- Temporary scaffold that fixes
 - Floating extrusions.
 - Long bridges.
 - Steep overhangs.
- Added by the slicer.
(Green trees-like branches.)
- Tree/organic supports can reach into difficult spaces.
- Extra complexity, plastic, and print time.

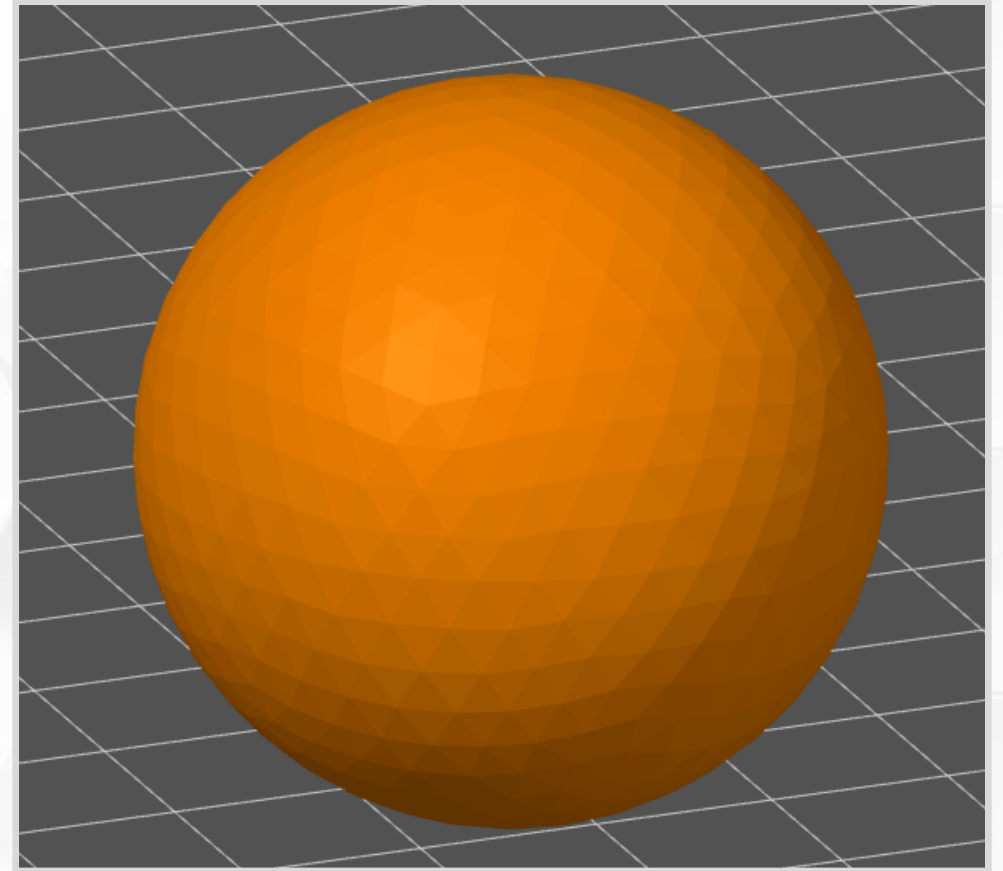


Supports

- In special situations, you can include supports as part of a 3D model.
 - *Example: 3D Printing Trick by Clough42, and a follow-up video.*
- **Supports are a precarious balance.** Why?
 - If they do not bond well enough, they hold nothing in place.
 - If they bond too well, they become part of the print.
- Supports can leave a rough surface finish.
- Sometimes you need to cut away pieces of support.

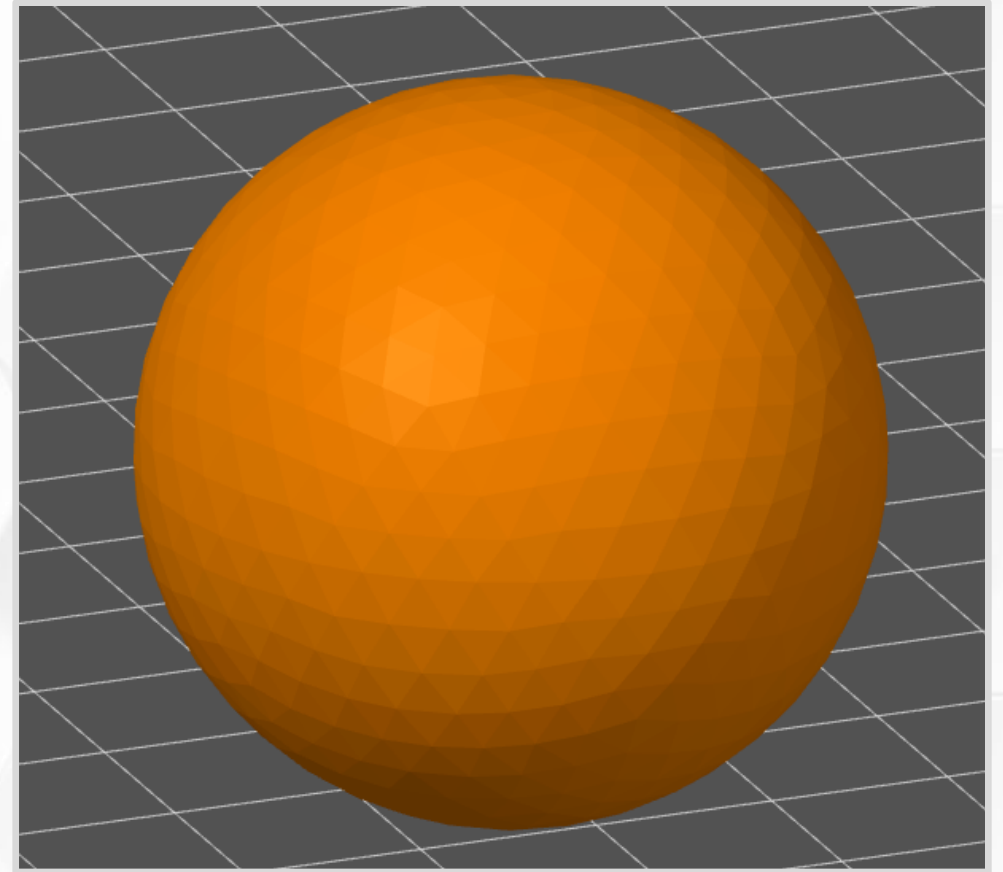
Limitations Review

- **Printability** encompasses all the limitations for a specific object.
- Without extra considerations, a simple sphere has poor printability.
- Why? Make a list of limitations.



Limitations Review

- In order of importance ...
- Very bad first layer contact. Significant spaghetti risk.
- Steep overhangs on the bottom.
- Second layer is steep enough to create floating extrusions.
- Uneven surface finish at the top and bottom-most layers.
- How can we fix this print?



Limitations Review

- Brim is not helpful.
 - First layer is too small.
- Add supports to the bottom.
- Cannot fix stair-stepping by changing print orientation.
 - Use small layer heights to reduce visual artifacts.
 - Use adaptive layer heights to make the surface look more uniform.
- Print two halves, then put them together.
 - Add tabs to help you align them.

Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

Thermal Expansion

Dimensional Accuracy

- Be careful modeling parts with exact dimensions.
- Why?
 - 3D prints are imperfect.
 - Errors compound when you assemble printed parts together.
- Where do these errors come from?
- And how do we fix it?



Dimensional Accuracy

- Sources of error include but are not limited to:
 - Filaments that shrink.
 - Shapes that easily warp.
 - Inconsistent extrusions.
 - Racked gantry (skewed axes).
 - Failing cooling fan.
 - Incorrect belt tension.
 - Backlash, slop, or loose parts.
 - Moving parts that are worn down.
 - Sloppy slicer settings.
 - Thermal expansion (*discussed in Part 3*).
 - Vibrations (*discussed in Part 3*).

Modeling Calibration

- $(\text{Printed size}) = (\text{Size}) \times (\text{Scale factor}) + (\text{Constant offset})$.
- Simplified model.
 - Assume fully functional motion system.
 - Apply to X and Y dimensions.
- **Scale factor.** Error changes with size.
 - Part shrinkage.
- **Constant offset.** Error is always the same.
 - Extrusion inconsistencies. Over and under-extruding.
 - Surface roughness. Wet filament, additives like carbon fiber, over-extruding.



(xkcd.com)

Fixing Scale Factor

- **Adjust part scaling in the slicer to fix scale factor.**
 - Scale XY size, or
 - Dedicated shrinkage compensation setting.
- *Example:* If you part shrinks by 1%, scale the size to 101%.
- When do you need part scaling?
 - Large parts over 6 inches, 150 mm.
 - Filaments that shrink, like ABS.
- I ignore scale factor for small projects that use PLA and PETG.

Fixing Constant Offset

- **Add clearance (air gap) between parts that touch to fix constant offset.**
- Very important for projects with assembled pieces!
 - All sizes.
 - All filaments.
- Best to add clearance while designing 3D objects.
 - Greatest flexibility. Independently adjust each surface.
- Or try adding negative XY offset in slicer.
 - Equally offsets all vertical surfaces.

Adding XY Clearance

- How much XY clearance? It depends ...
- **Start with 0.25 mm.** Offset each side by half, 0.125 mm.
- *Example:* Making a hole for a bolt.
 - Add 0.2 - 0.3 mm of clearance for a close fit.
- *Example:* Making a lid that easily slides into place.
 - Add 0.4 - 0.8 mm of clearance for a loose fit.

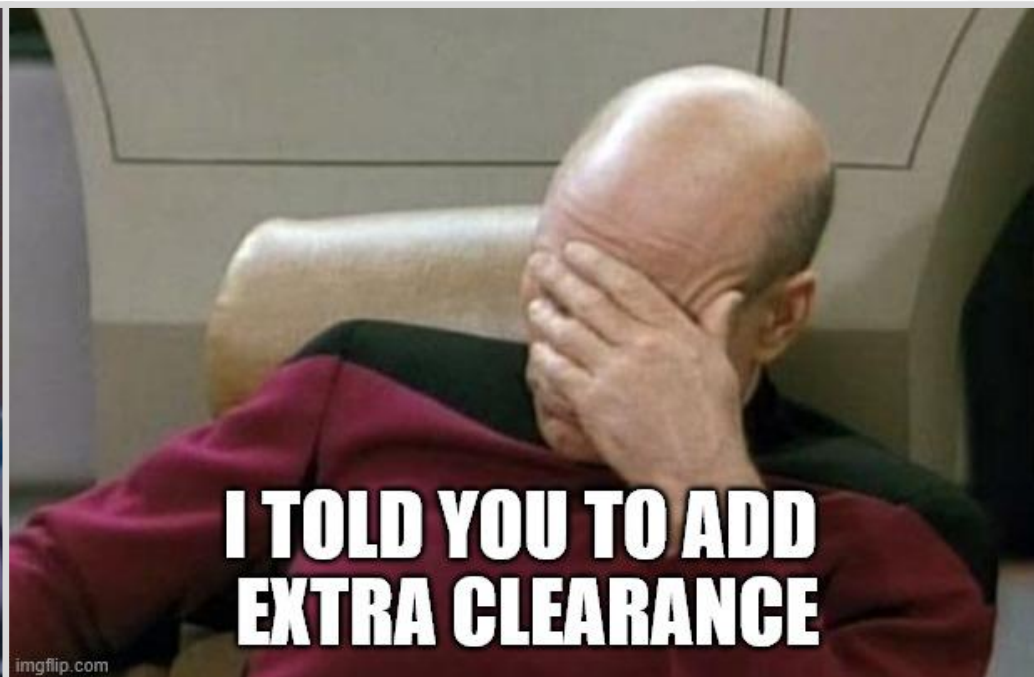
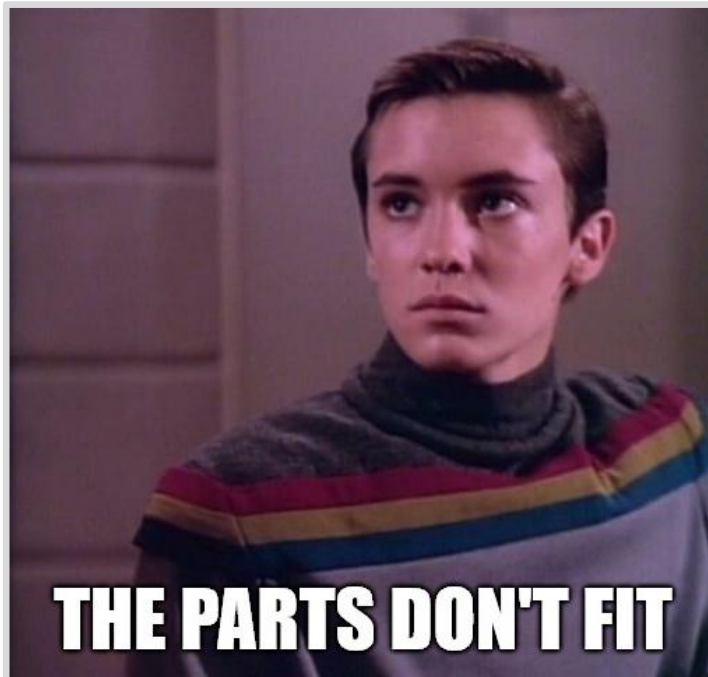
Adding Z Clearance

- How much Z clearance? It depends ...
- **Start with one layer or 0.1 mm.**
- When should you increase clearance?
 - Sagging extrusions.
 - Rough bottom surface from supports.
 - High and low spots on print bed.
 - Tilted print bed.

Summary

If your project has assembled parts then ...

- You sometimes need scale factor.
- You **always** need constant offset.



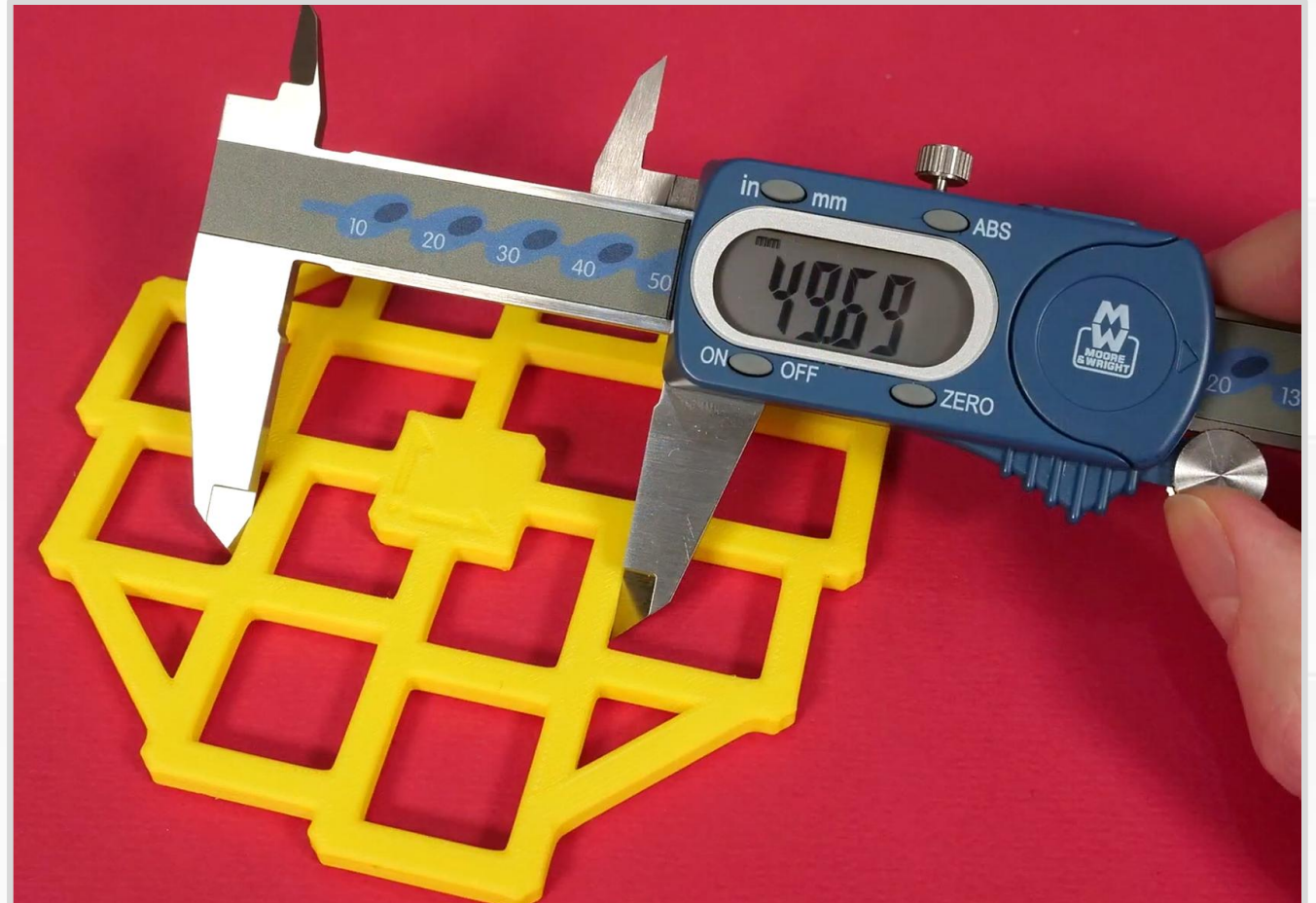
Calibration Prints

- Calibration cubes.
 - Labels for X and Y-axis.
 - Tells you constant offset for outside dimensions.
 - Simple but still useful.
- Clearance test by 3DMakerNoob.
 - Print-in-place clearance/tolerance.
 - Alters the gap around discs.
 - Feel which pieces move freely.



Calibration Prints

- Califlower by Vector3D.
 - Paid model.
 - Includes walkthrough, code, and calculator.
 - Inner dimensions.
 - Outer dimensions.
 - Shrinkage.
 - Skew (if axes are not 90°).
- Calibration video by CNC Kitchen.



Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

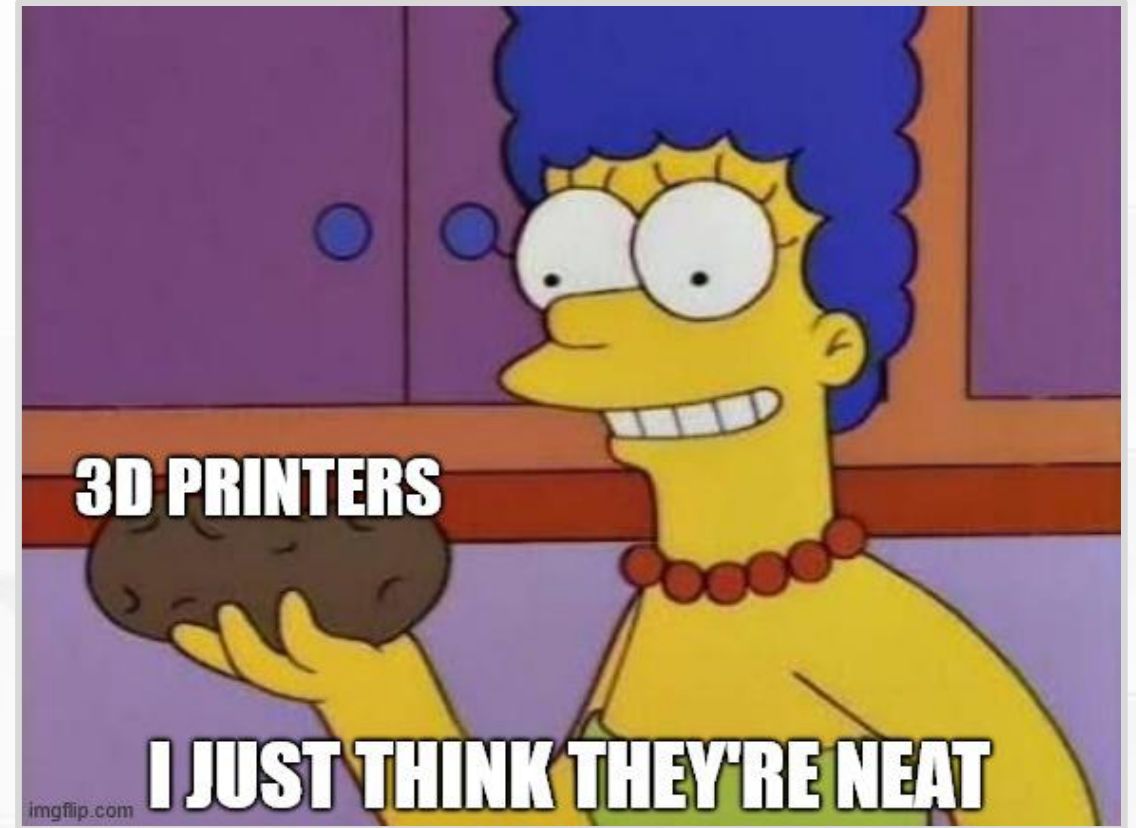
Vibrations and Waves

Linear Motion

Thermal Expansion

An Amazing Tool

- Playground for **creative design** and **creative problem solving**.
- Massive utility for cheap.
- Diverse applications.
 - Art projects, toys, DIY and repair, tools for work and research, etc.
- Bring your ideas to life.
- Fast turnaround time.
- Relatively low barrier to entry.

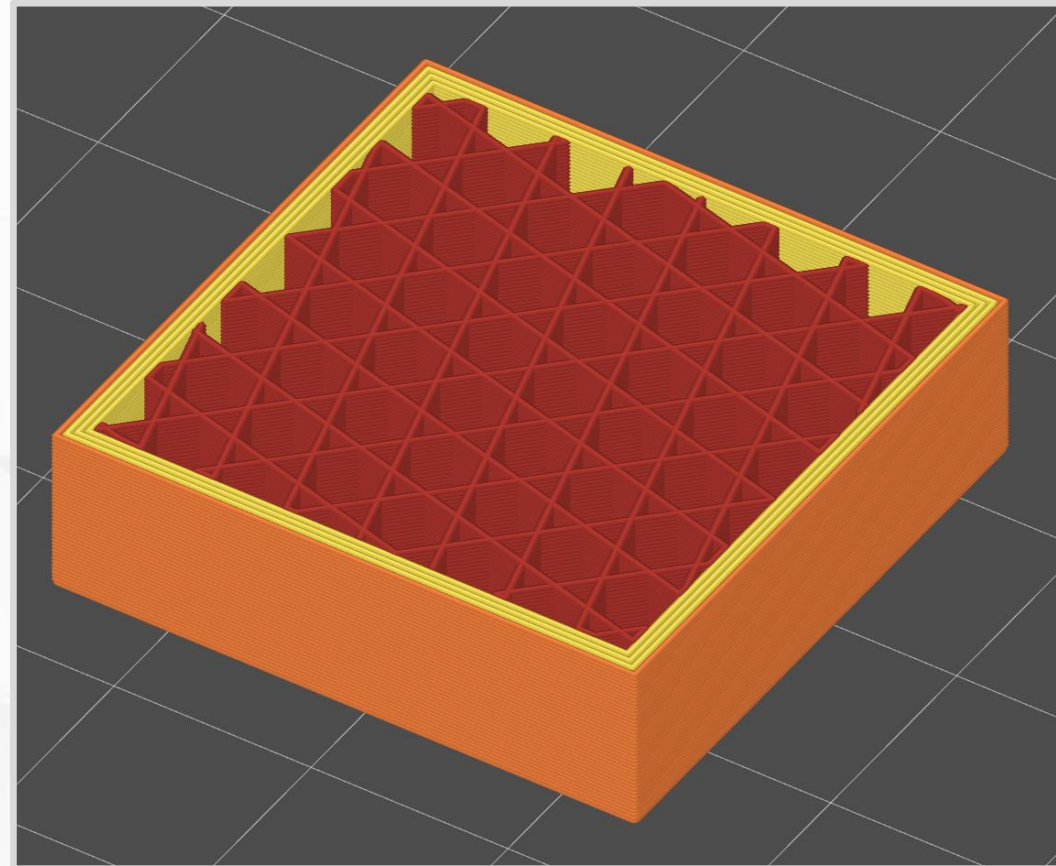


An Amazing Community

- RepRap festivals, maker fairs, local clubs and maker spaces, ...
- Projects that make a difference. Examples include:
Toddler mobility trainer by MakeGood
- People share cool projects and ideas. Examples include:
Full spectrum printing
BumpMesh.com
Voron
Countless free, downloadable models
- YouTube makers with educational content. Examples include:
CNC Kitchen
Maker's Muse
JanTec Engineering
3D Printing Nerd
Teaching Tech (retired)

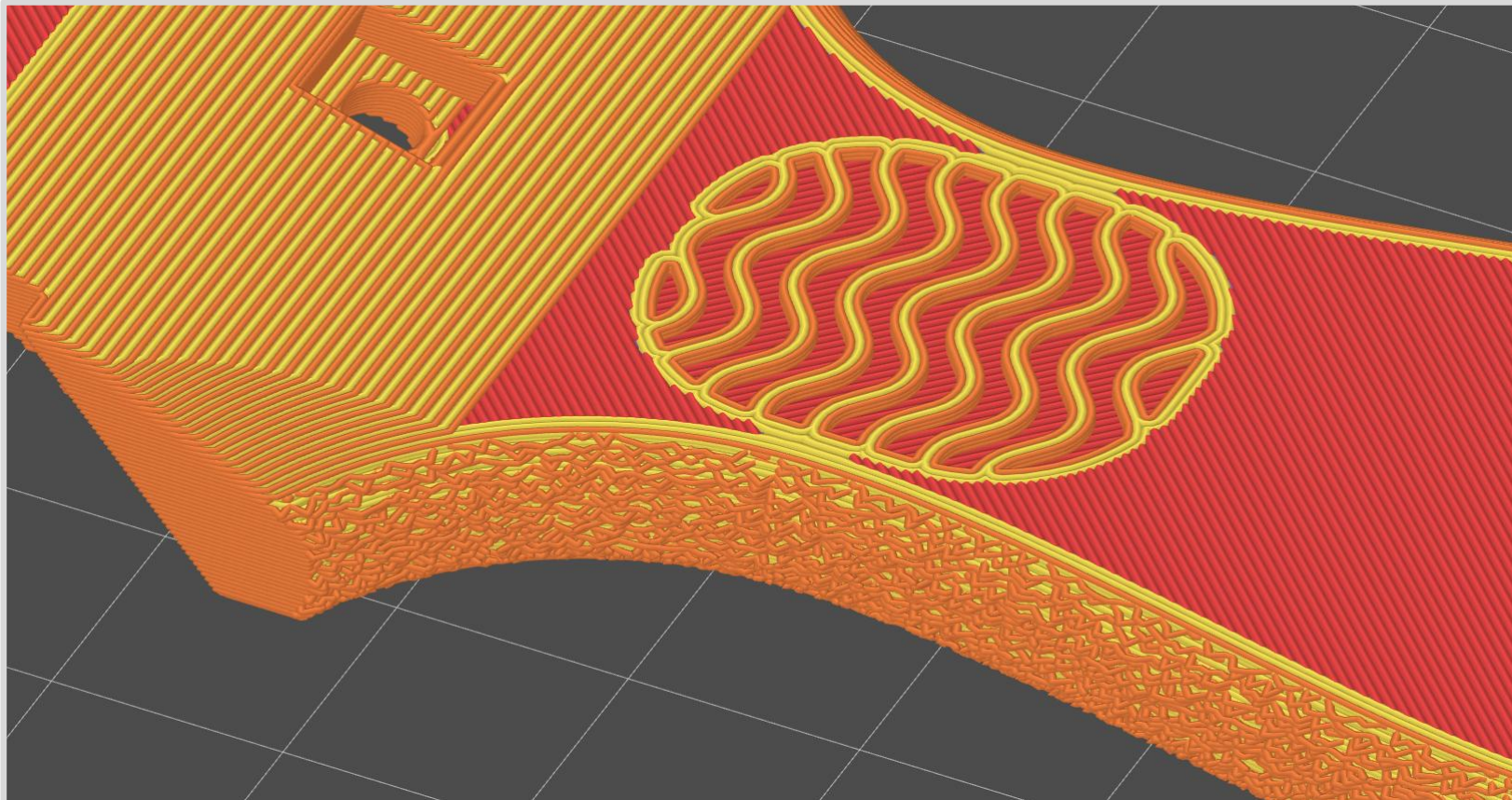
FFF Advantages

- Make lots of shapes with tiny “building blocks.”
 - Typical extrusions are **0.2 mm tall** (1-2 pieces of paper) and **0.4 mm wide**.
- Not restricted to uniform, solid objects.
 - Easily create internal geometry.
 - Great at making hollow objects.



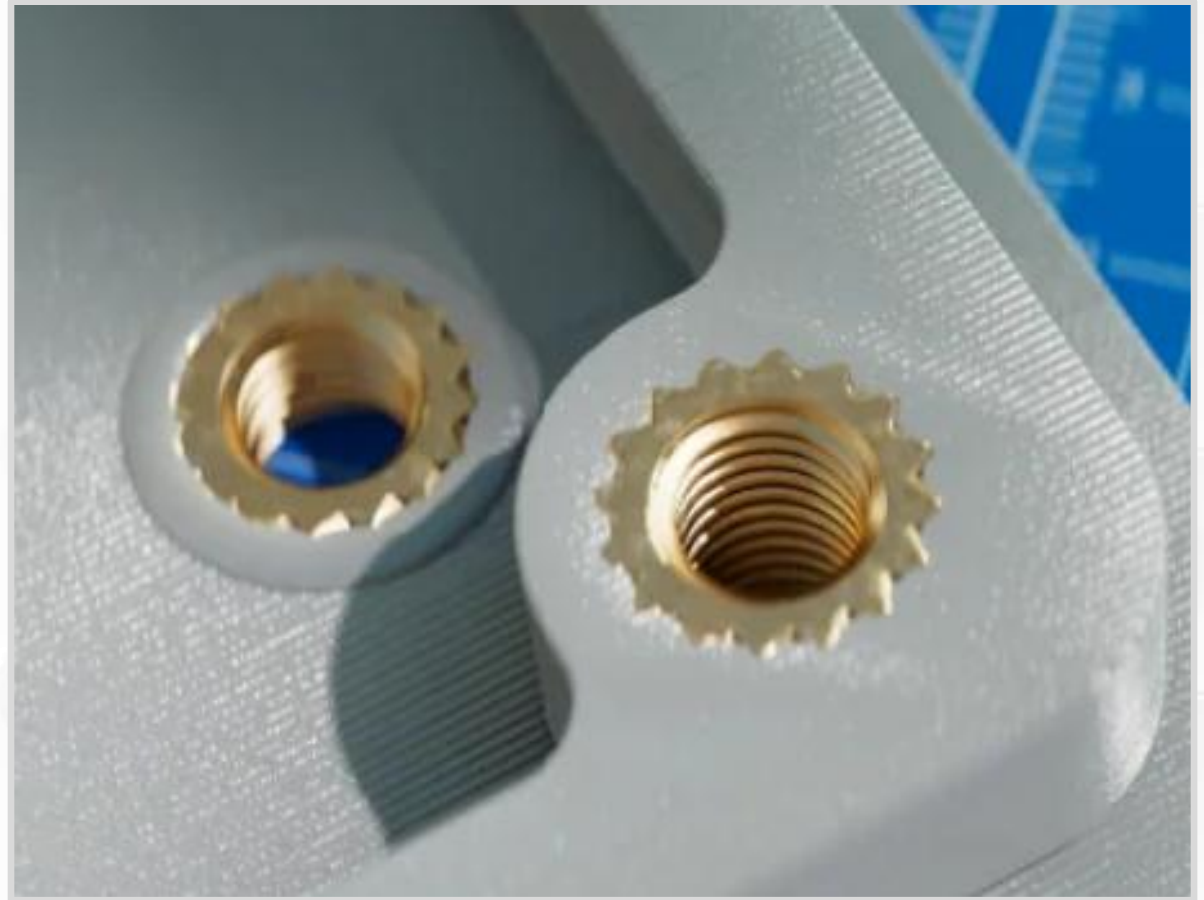
Custom Slicing

- Change slicer settings for specific regions or surfaces. Too many to list.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



Heat Set Inserts

- Threaded anchor for screws. Why?
- Plastic is soft → Screws easily wear out plastic holes.
- Install with a hot soldering iron.
- Locks in place when it cools.
- Very durable.



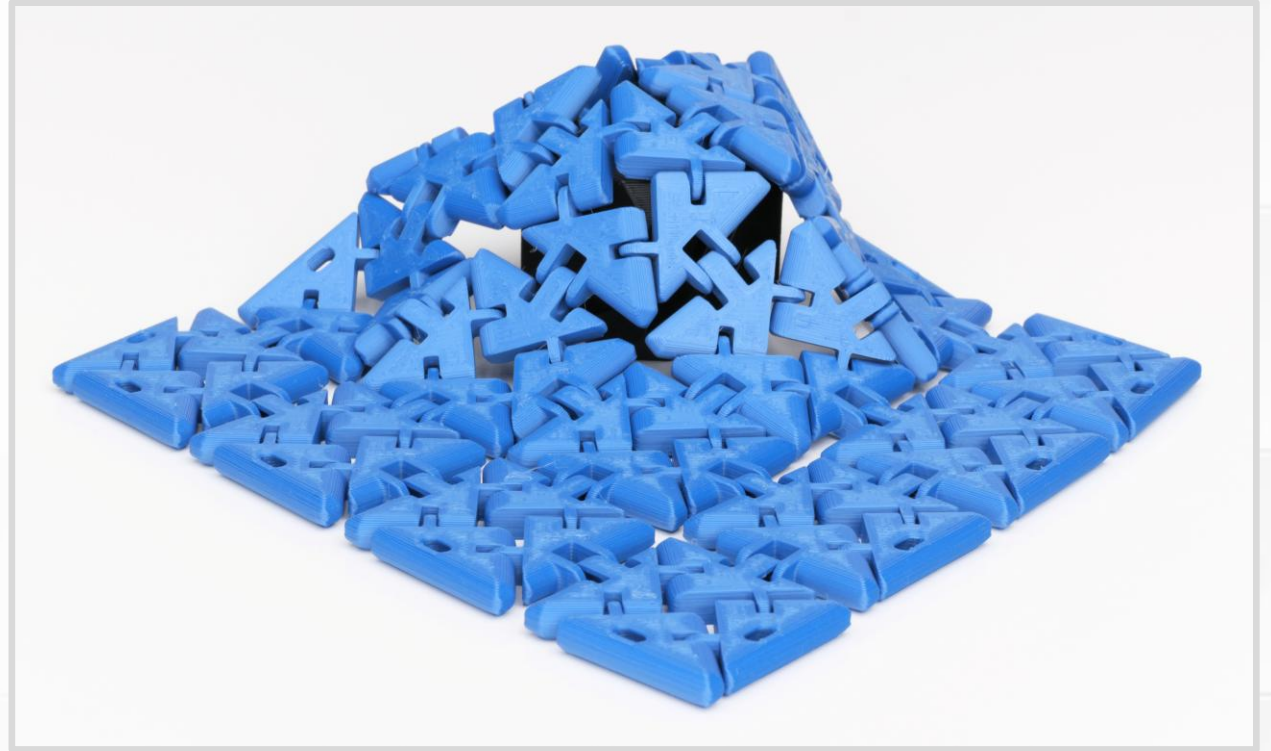
(Pictures from [CNC Kitchen Store](#))

Captive Parts

- Insert an object into a pocket or void in the middle of a print.
 - Nuts and fasteners.
 - Magnets.
 - Weights.
 - Reinforcing bar or plate.
- Locked inside when the print finishes.
- Object **must** sit below the current layer, or it creates a collision with the toolhead.
- A little inconvenient.
 - Need to be at the printer at the right layer.

Print-in-Place (PIP)

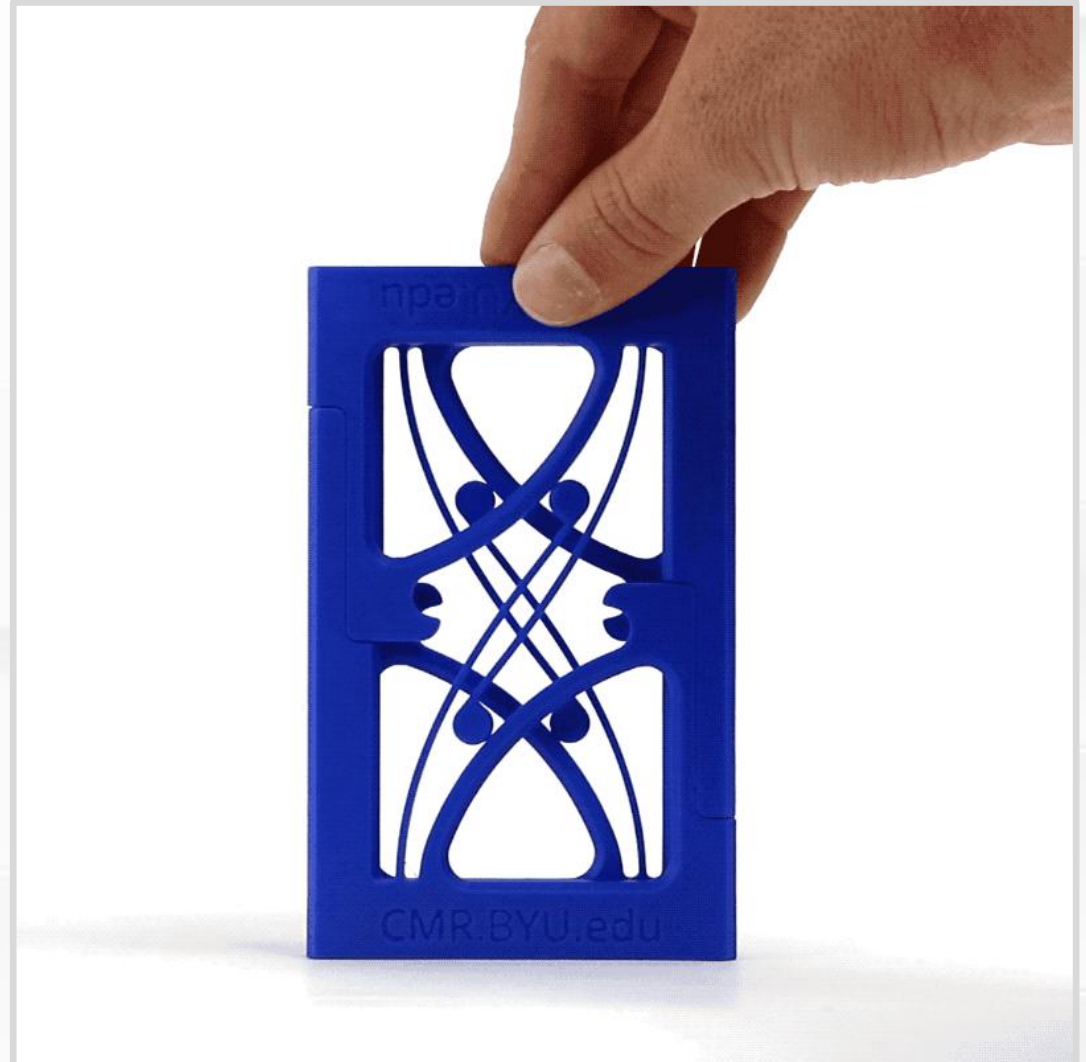
- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by FlyingGyroscope



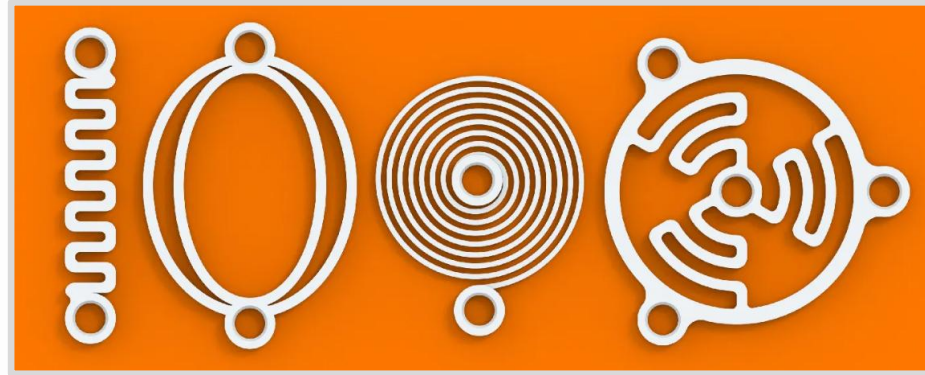
Functional Flexing

- Springs and **compliant mechanisms**.
- Features and shapes that function by bending and flexing.
- Elastic, springy materials.
 - No permanent deformation.
- Remember part strength and layer lines.

(Animation, Constant Force
Mechanism by BYU CMR)



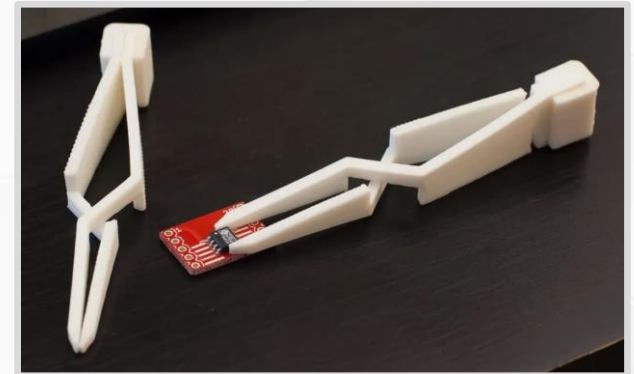
Springs by VC Design



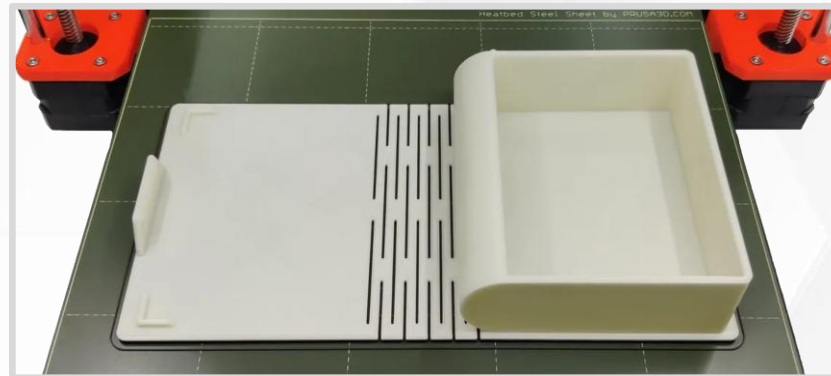
Clip by AE



Cross tweezer by joo



Living hinge by DrJones





github.com/FlyingGyroscope/IntroTo3DPrinting/